Fundamentals of Information System

Introduction

Course Aim:

❖ **Students will learn to :**

- Understand how and why information systems are used today.
- Explain the technology, people, and organizational components of information systems.
- Understand globalization and the role information systems has played in this evolution.
- Understand how businesses are using information systems for competitive advantage vs. competitive necessity.
- Understand the value of information systems investments as well as learn to formulate a business case for a new information system, including estimation of both costs and benefits.
- Know the major components of an information systems infrastructure.
- Mitigate risks as well as plan for and recover from disasters.
- Understand how information systems are enabling new forms of commerce between individuals, organizations, and governments.
- Be aware of emerging technologies that enable new forms of communication, collaboration, and partnering.

Content:

- An introduction to information systems and their foundational technologies.
- Topics include surveys of hardware, operating systems, database and networked computing concepts,
- as well as special topics in information systems including computation for the internet and world wide web, distributed computing, electronic commerce systems, artificial intelligence, decision support systems, security technologies, and information systems design.
- Themes common to each of these topics include security and privacy issues, ethics, and cost/benefit analyses.

Main Text book:

- 1. Information Systems Technology, Ross A. Malaga, ©2005 Prentice Hall.

Grading

- **Assignments & Homework + Midterm Exam** **30 %**
- **Attendance** **10 %**
- **Final Exam** **60%**

Information - Definition

❖The word information is very vague and unclear in its common usage.

❖But for the purpose of this subject

- Information is data that has been processed into a form that is meaningful to the recipient and is of real or perceived value in current or prospective actions.

Principles and Learning Objectives

- The value of information is directly linked to how it helps decision makers achieve the organization's goals
  - Distinguish data from information and describe the characteristics used to evaluate the quality of data

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Cont…

- Knowing the potential impact of information systems and having the ability to put this knowledge to work can result in a successful personal career, organizations that reach their goals, and a society with a higher quality of life

  – Identify the basic types of business information systems and discuss who uses them, how they are used, and what kinds of benefits they deliver

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Cont…

- System users, business managers, and information systems professionals must work together to build a successful information system
  - Identify the major steps of the systems development process and state the goal of each

- Information systems personnel are the key to unlocking the potential of any new or modified system

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Cont…

- IS personnel is a key to unlocking the potential of any new or modified system

– Define the types of roles, functions, and careers available in information systems

Why Learn About Information Systems in Organizations?

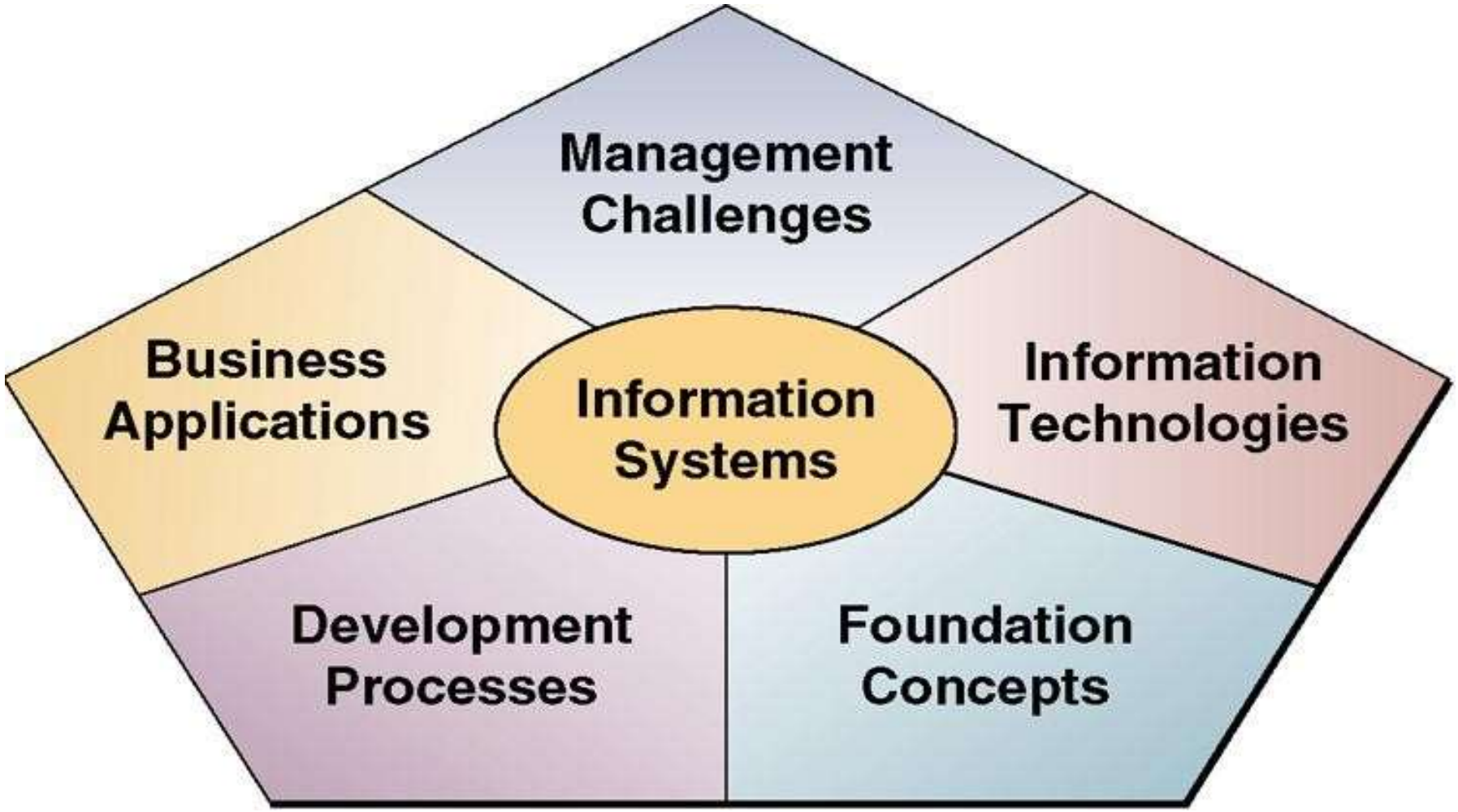
❖ Information systems used by almost every imaginable profession

- Entrepreneurs and small business owners
- Sales representatives
- Managers
- Financial advisors

❖Information systems:

- Indispensable tools to help you achieve your career goals

An IS Framework for Business Professionals



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Cont…

- Five areas of knowledge are important to end users in order to understand information systems:

- **Foundation Concepts.**

  - End users must be familiar with the basic components and types of information systems there are.

  - But they also need to be familiar with general systems theory and theories of information processing (machine and human).

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❖ Information Technology.

- End users should understand technology, more precisely, the information technology of hardware, software, telecommunications, database management, and how all these elements interaction in a dynamic process of very rapid change, development, and new ways of doing business

❖ Business Applications.

- How information systems are applied to business problems is more complex than it might seem.

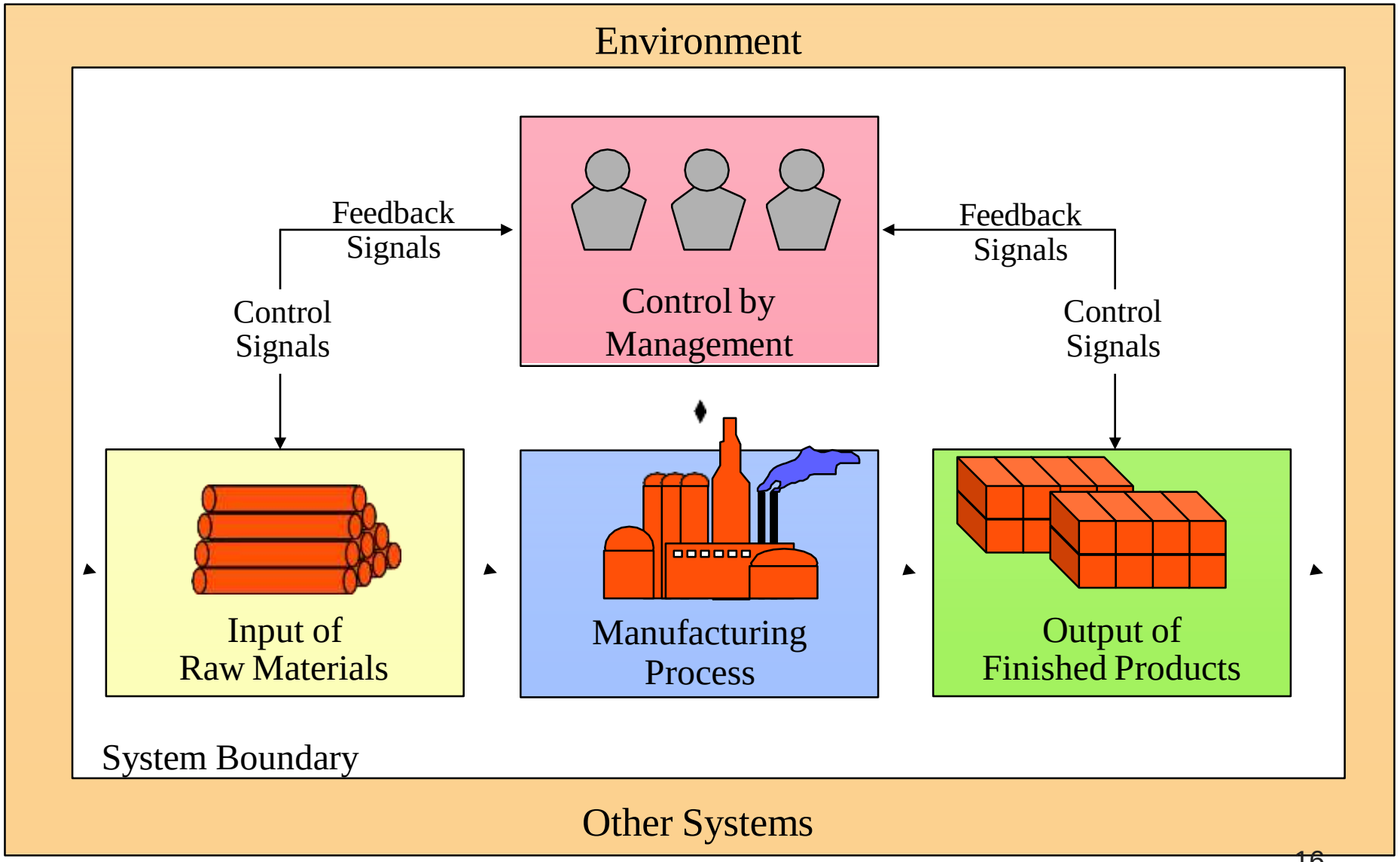
- The informed end users seeks to learn both about how to use information systems to solve existing problems and to begin using IS as a new way of defining problems and meeting business opportunities.

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- **Development Processes**.
  - End users of IS need to know the fundamental concepts of problem-solving and development.
  - Here you should become familiar with methodologies such as the systems approach, the systems development life cycle, and prototyping
- **Management Challenges**.
  - How managers make use of IS resources is a key concern for end users.
  - More than ever, a knowledge of management methods is required by each end users, as IT demands that end users make more independent decisions that support the company's overall objectives.

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What is a System?



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System Concepts

- **SYSTEM** is a group of interrelated components working together toward a common goal by accepting inputs and producing outputs in an organized transformation process. Systems have the following three basic interacting functions.

  - **Input**.
  - **Processing**.
  - **Output**.

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What the meaning of IS?

- **Information system (IS)**
  - Set of interrelated components that collect, manipulate, store, and disseminate data and information and Provides feedback to meet an objective
  - **Examples**: ATMs, airline reservation systems, course reservation systems
- **Businesses:**
  - Can use information systems to increase revenues, improve customer service and reduce costs

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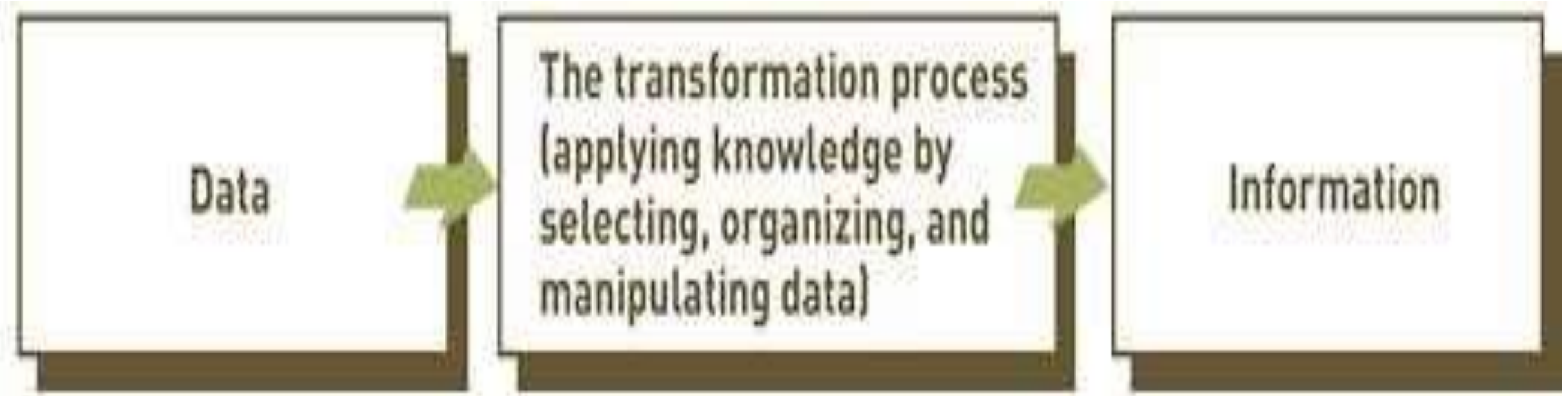
Data, Information and Knowledge

- **Data:**
  - Raw facts
- **Information:**
  - Collection of facts organized and processed in such a way that they have value beyond the individual facts
- **Process:**
  - Set of logically related tasks performed to achieve a defined outcome
- **Knowledge:**
  - Awareness and understanding of a set of information and the ways information can be made useful

Types of Data

Data	Represented by
Alphanumeric data	Numbers, letters, and other characters
Image data	Graphic images and pictures
Audio data	Sound, noise, or tones
Video data	Moving images or pictures

Cont…



The Process of Transforming Data into Information

The Value of Information

- Directly linked to how it helps decision makers achieve their organization’s goals Valuable information:
- Can help people and their organizations perform tasks more **efficiently** and **effectively**
- Inaccurate data can result in loss of potential new customers and reduce customer satisfaction

Table 1.2

Characteristics of Valuable Information

Characteristics	Definitions
Accessible	Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs.
Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process. (This is commonly called garbage in, garbage out [GIGO].)
Complete	Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it.
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the total value the company has invested in inventory.
Relevant	Relevant information is important to the decision maker. Information showing that lumber prices might drop might not be relevant to a computer chip manufacturer.
Reliable	Reliable information can be trusted by users. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up might not be reliable.
Secure	Information should be secure from access by unauthorized users.
Simple	Information should be simple, not overly complex. Sophisticated and detailed information might not be needed. In fact, too much information can cause information overload, whereby a decision maker has too much information and is unable to determine what is really important.
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.

What Is an Information System?

❖**Information system (IS) is a set of interrelated elements that:**

- Collect (input)
- Manipulate (process)
- Store
- Disseminate (output) data and information
- Provide a corrective reaction (feedback mechanism) to meet an objective

Input, Processing, Output, Feedback

❖Input:

- Activity of gathering and capturing raw data

❖Processing:

- Converting data into useful outputs

❖Output:

- Producing useful information

❖Feedback:

- Information from the system that is used to make changes to input or processing activities

Computer-Based Information Systems

- Single set of hardware, software, databases, telecommunications, people, and procedures:
  - That are configured to collect, manipulate, store, and process data into information
- **Technology infrastructure:**
  - Includes all hardware, software, databases, telecommunications, people, and procedures configured to collect, manipulate, store, and process data into information

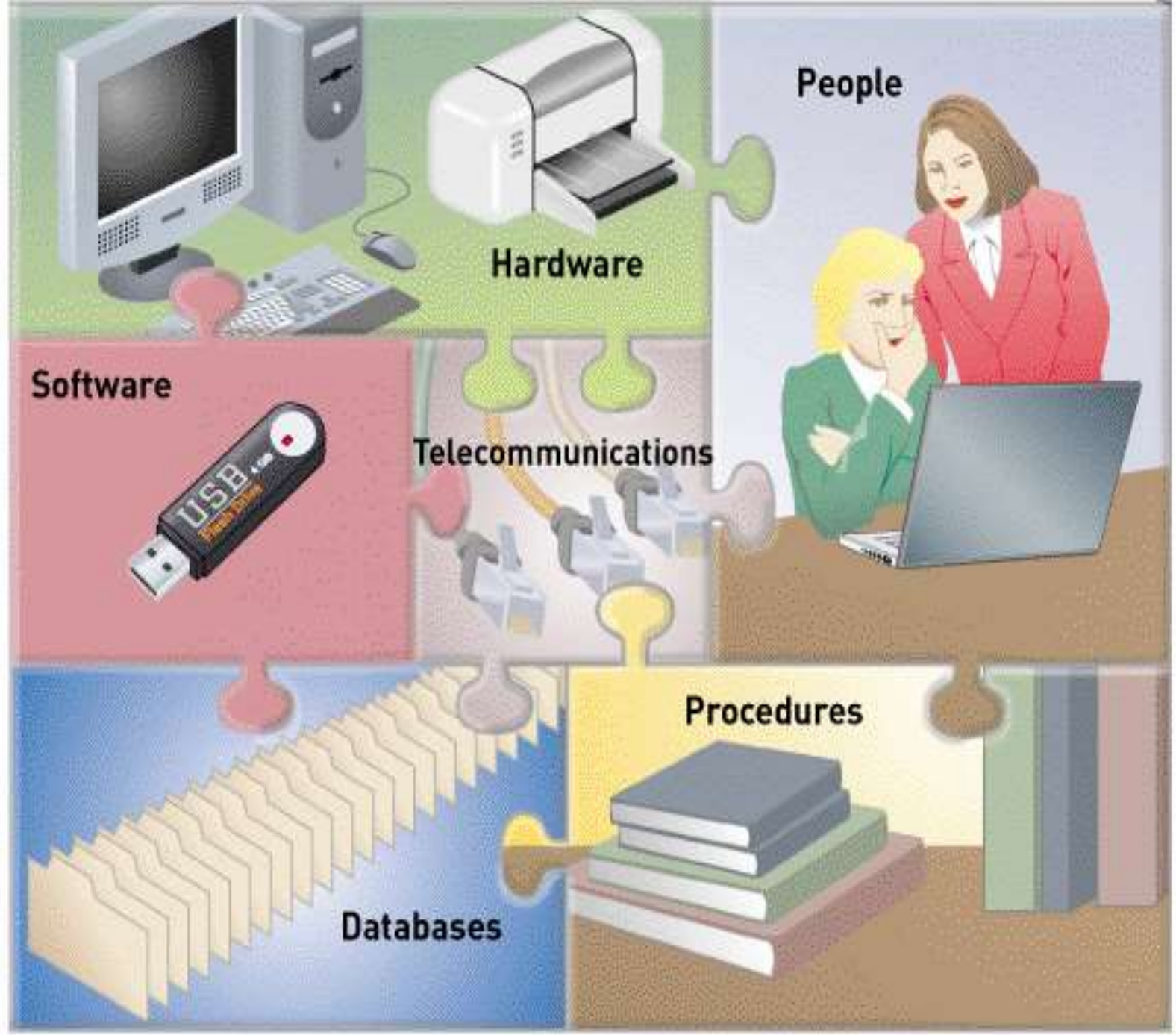


FIGURE **1.4**

Components of a computer-based information system

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Computer-Based Information Systems (continued)

- **Hardware:**
  - Computer equipment used to perform input, processing, storage, and output activities
- **Software:**
  - Computer programs that govern the operation of the computer
    - System software controls basic computer operations
    - Application software allows you to accomplish specific tasks
- **Database:**
  - Organized collection of facts and information, typically consisting of two or more related data files

Computer-Based Information Systems (continued)

- **Telecommunications, networks, and the Internet:**
  - The electronic transmission of signals for communications
- **Networks:**
  - Connect computers and equipment to enable electronic communication
- **Internet:**
  - World's largest computer network, consisting of thousands of interconnected networks, all freely exchanging information

Computer-Based Information Systems (continued)

- **Web:**
  - Network of links on the Internet to documents containing text, graphics, video, and sound
- **Intranet:**
  - Internal network that allows people within an organization to exchange information and work on projects
- **Extranet:**
  - Network that allows selected outsiders, such as business partners and customers, to access authorized resources of a company's intranet

Computer-Based Information Systems (continued)

- **People:**
  - The most important element in most computer based information systems
- **Procedures:**
  - Include strategies, policies, methods, and rules for using the CBIS

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Thank you for your ATTENTION

Fundamentals of Information System

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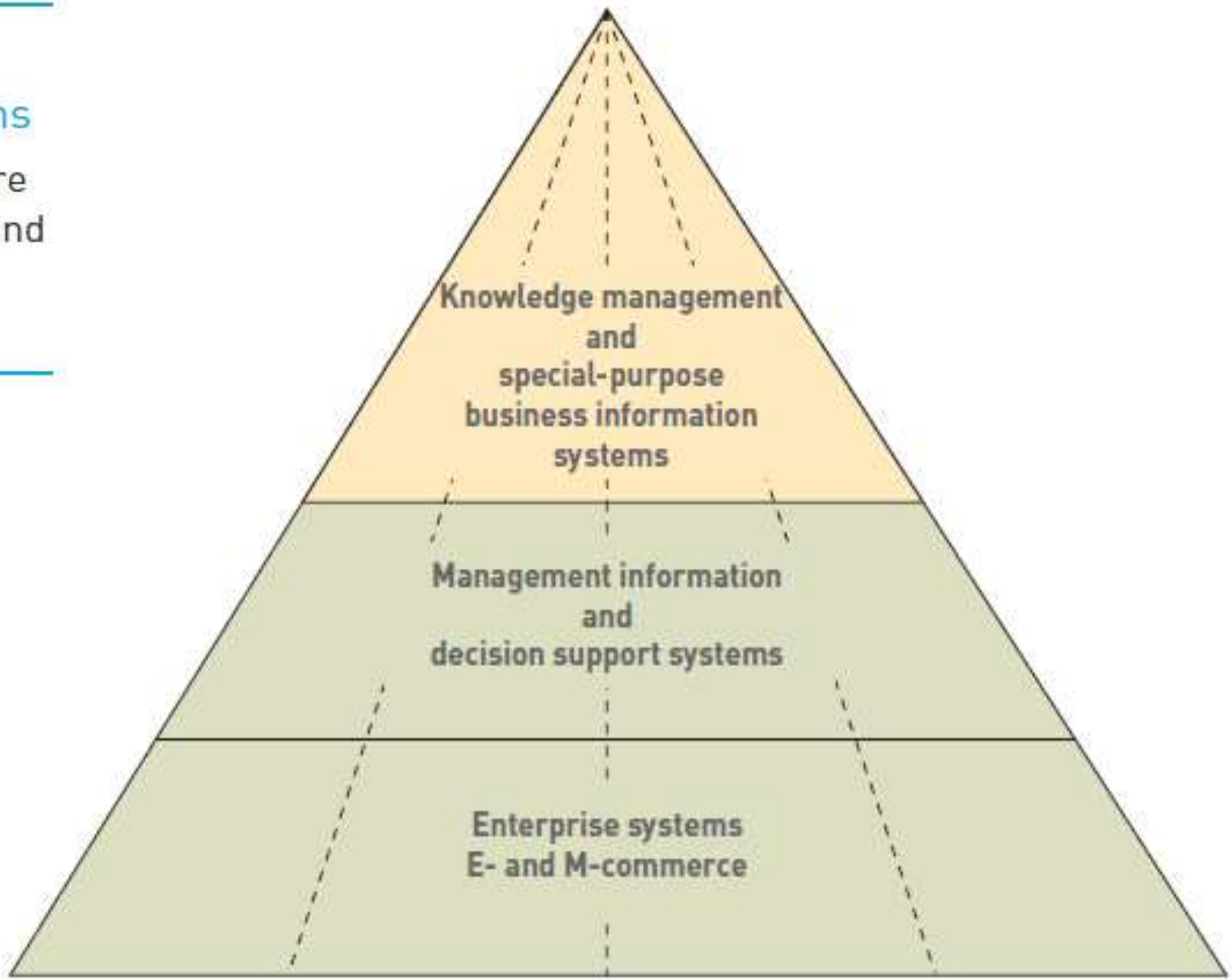
Business Information Systems

- **Most common types of information systems:**
  - Those designed for electronic and mobile commerce, transaction processing, management information, and decision support
- **Some organizations employ:**
  - Special-purpose systems, such as virtual reality, that not every organization uses

Figure 1.5

Business Information Systems

Business information systems are often integrated in one product and can be delivered by the same software package.



Electronic and Mobile Commerce

- **E-commerce:**
  - **Any business transaction executed electronically between:**
    - Companies (business-to-business, B2B)
    - Companies and consumers (business-to-consumer, B2C)
    - Consumers and other consumers (consumer- to consumer, C2C)
    - Business and the public sector
    - Consumers and the public sector

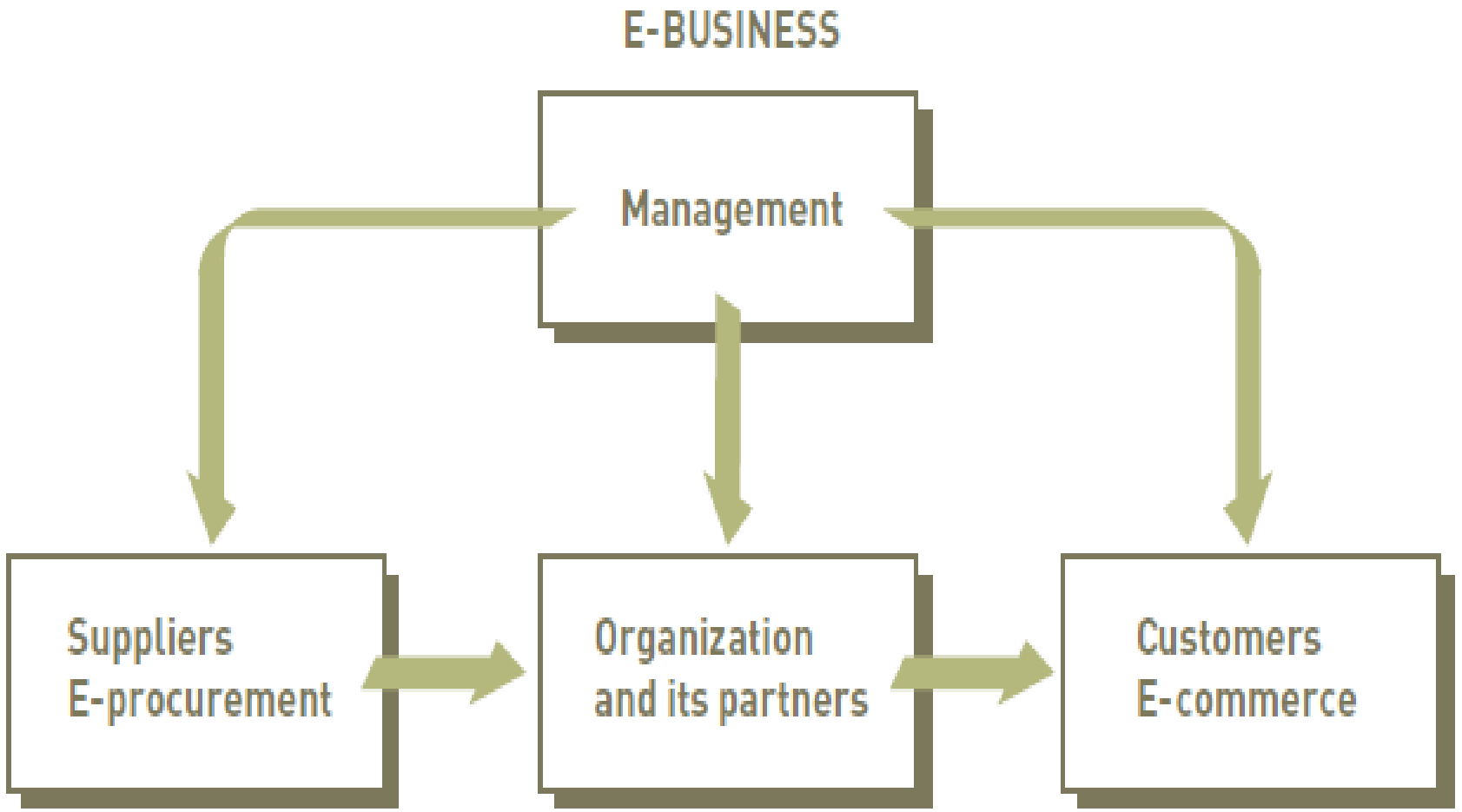
Electronic and Mobile Commerce (continued)

- **Mobile commerce (m-commerce):**
  - The use of mobile, wireless devices to place orders and conduct business
- **E-commerce:**
  - Offers many opportunities for streamlining work activities
- **Electronic business (e-business):**
  - Uses **information systems** and the **Internet** to perform all business-related tasks and functions

Figure 1.8

Electronic Business

E-business goes beyond e-commerce to include using information systems and the Internet to perform all business-related tasks and functions, such as accounting, finance, marketing, manufacturing, and human resources activities.



Enterprise Systems:

- **Transaction Processing Systems and Enterprise Resource Planning**
  - **Transaction:**
    - Any business-related exchange, such as payments to employees and sales to customers
  - **Transaction processing system (TPS):**
    - Organized collection of people, procedures, software, databases, and devices used t**o perform and record completed business transactions**

Enterprise Systems:

- **Enterprise Resource Planning: (ERP)**
  - Set of integrated programs that manages the vital business operations for an entire multisite, global organization

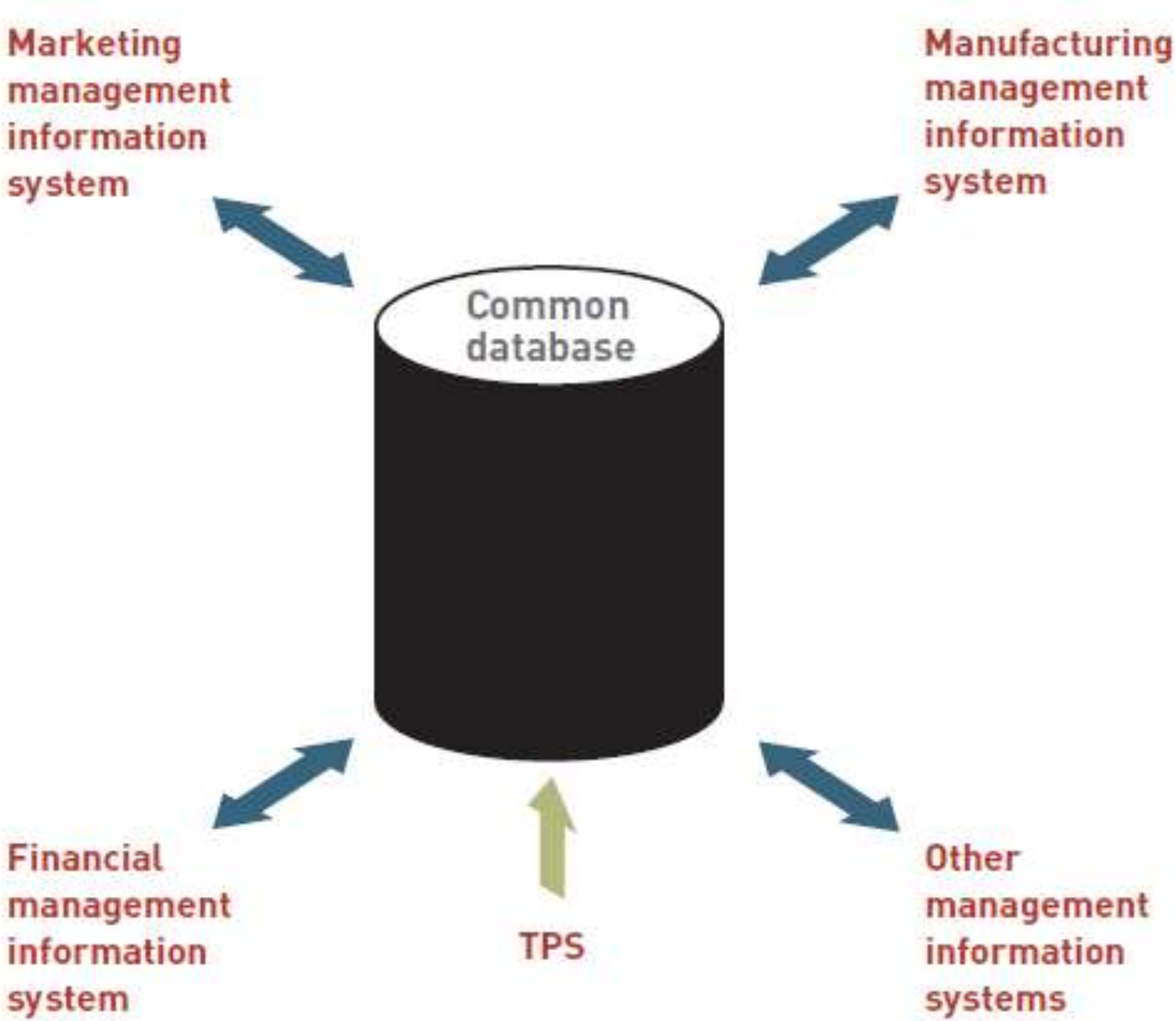
Information and Decision Support Systems

- **Management information system (MIS):**
  - Organized collection of people, procedures, software, databases, and devices **that provides routine information to managers and decision makers**
  - Manufacturing, marketing, production, finance, and other functional areas are supported by MISs and are linked through a common database

Figure 1.10

Management Information System

Functional management information systems draw data from the organization's transaction processing system.

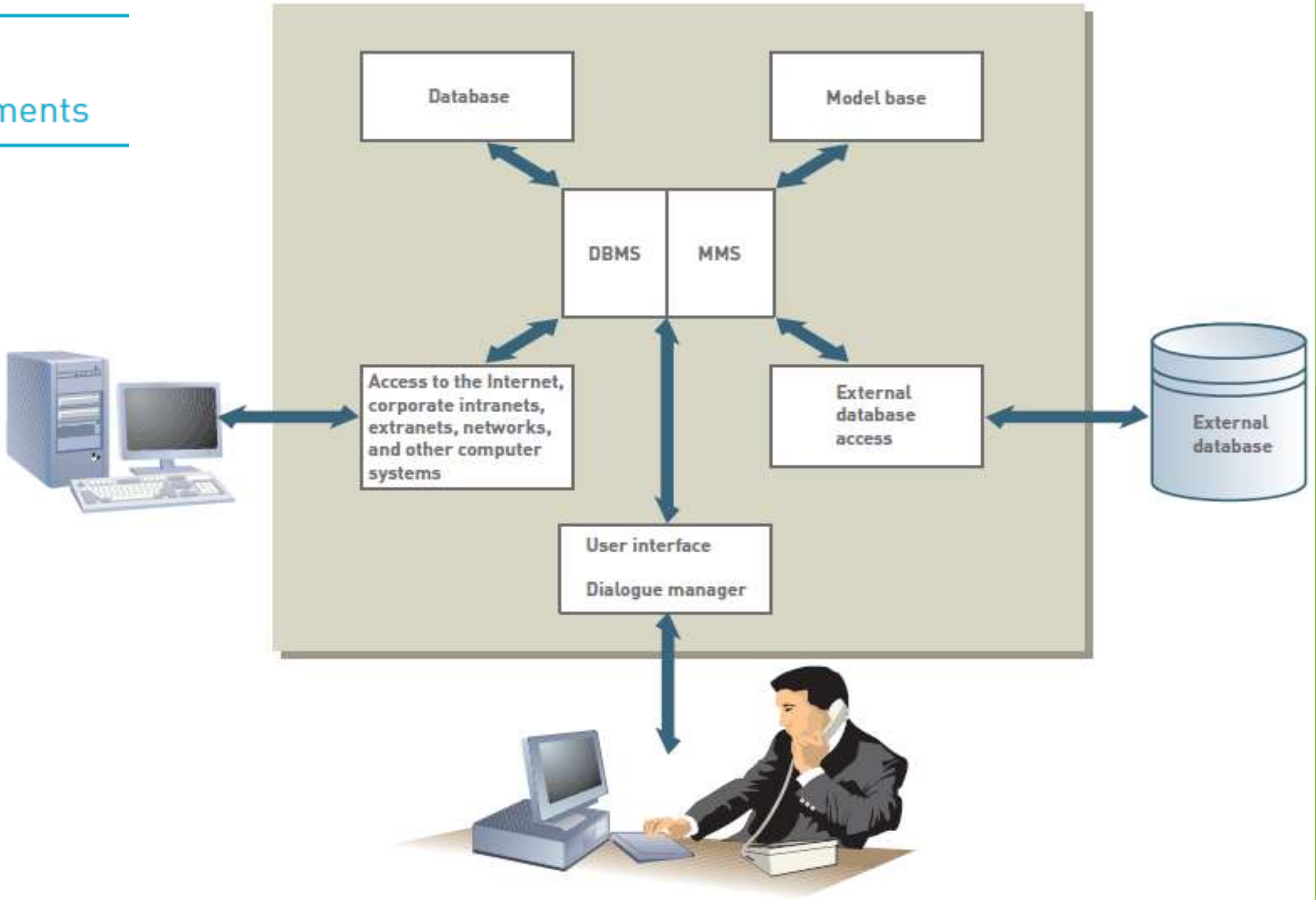


Information and Decision Support Systems

- **Decision support system (DSS):**
  - Organized collection of people, procedures, software, databases, and devices that **support problem-specific decision making**
  - **Can include:**
    - A collection of models used to support a decision maker or user (model base)
    - A collection of facts and information to assist in decision making (database)
    - Systems and procedures (user interface or dialogue manager) that help decision makers and other users interact with the DSS

Figure 1.11

Essential DSS Elements



Specialized Business Information Systems:

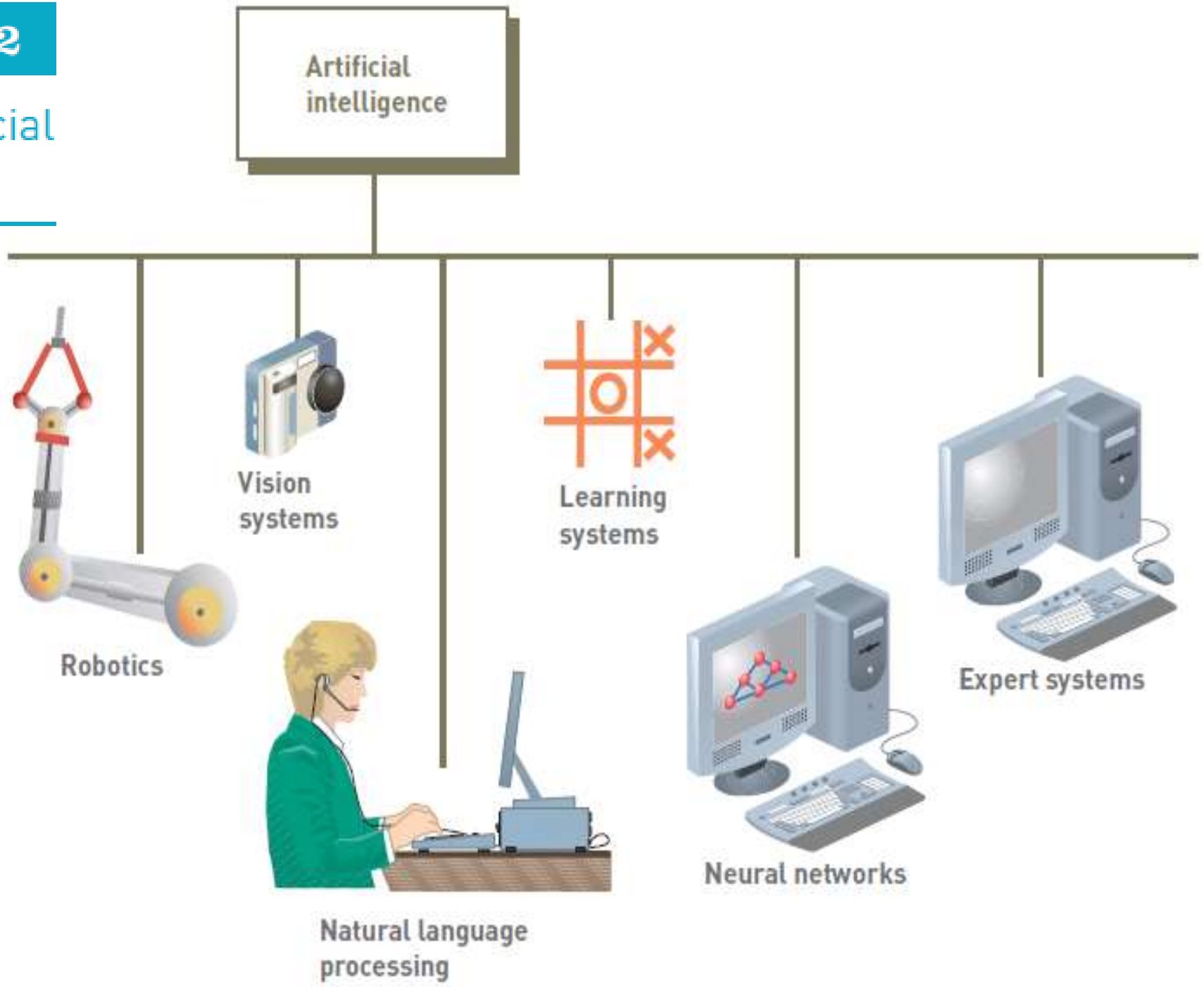
- **Knowledge Management Systems (KMS):**
  - Organized collection of people, procedures, software, databases, and devices to:
    - Create, store, share, and use the organization's knowledge and experience
- **Artificial Intelligence (AI):**
  - Computer system takes on characteristics of human intelligence

Specialized Business Information Systems:

- **Artificial intelligence (AI):**
  - Computer system takes on characteristics of human intelligence
  - Robotics and nanobots
  - Vision systems
  - Natural language processing
  - Learning systems
  - Neural networks

Figure 1.12

The Major Elements of Artificial Intelligence



Specialized Business Information Systems:

- **Expert systems:**
  - Give computer ability to make suggestions and function like an expert in a particular field
- **Virtual reality and multimedia:**
- **Virtual reality:**
  - Simulation of a real or imagined environment that can be experienced visually in three dimensions
  - Augmented reality superimposes digital data over photos or images
- **Multimedia:**
  - Can include photos and images, the manipulation of sound, and special 3D effects

PART 2

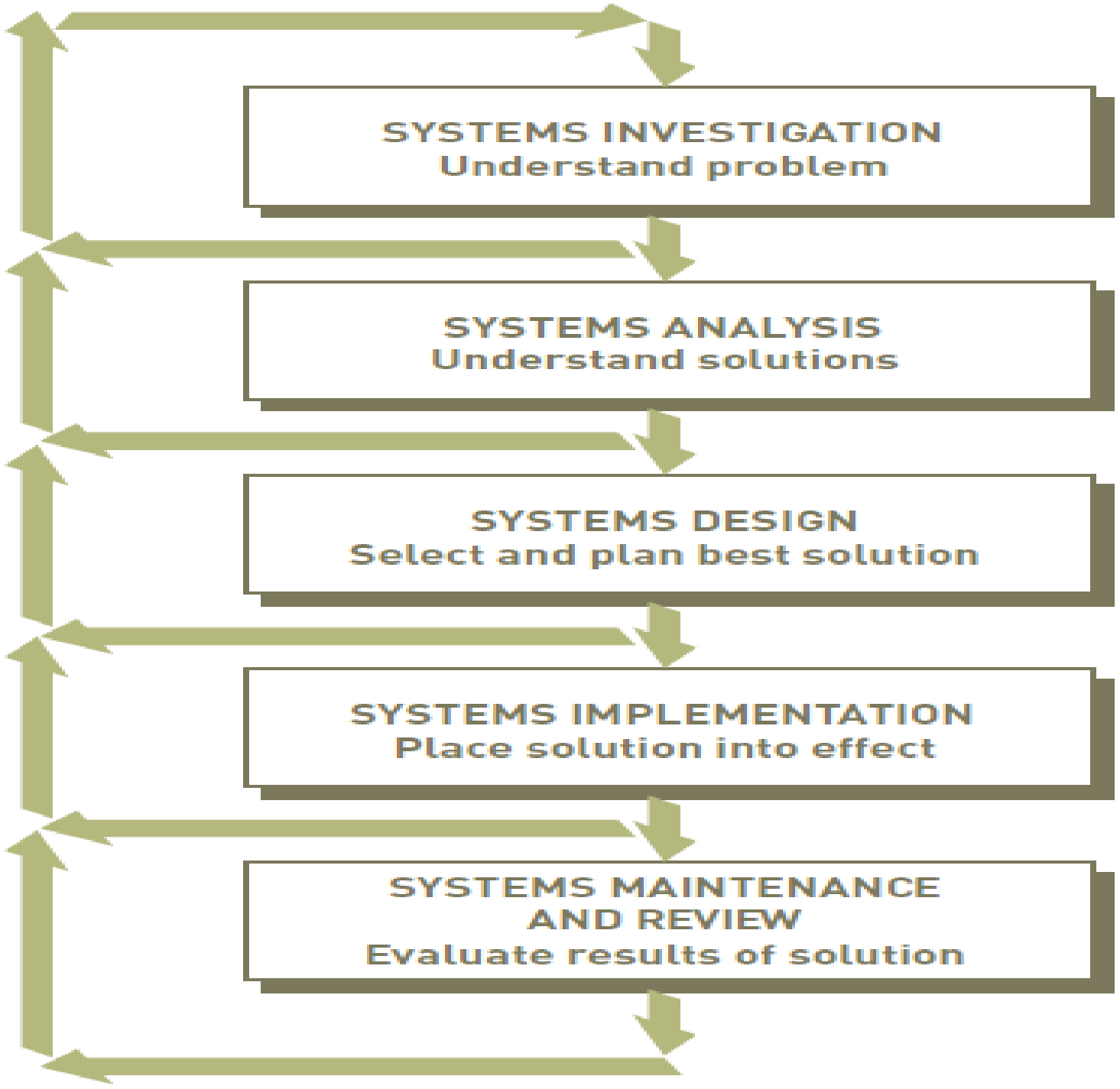
Systems Development

Systems Development

- **Systems development:**
  - The activity of creating or modifying existing business systems
- **Outsourcing:**
  - Allows a company to focus on what it does best and delegate other functions to companies with expertise in systems development
  - System development often outsourced

Figure 1.14

An Overview of Systems Development



Organizations and Information Systems

- **Organization:**
  - Formal collection of people and other resources established to accomplish a set of goals
  - Constantly uses money, people, materials, machines and other equipment, data, information, and decisions
  - Providing value to a stakeholder is the primary goal of any organization

Organizations and Information Systems

- **Value chain:**
  - Series (chain) of activities that includes inbound logistics and warehouse and storage, production and manufacturing, finished product storage, outbound logistics, marketing and sales, and customer service

Organizations and Information Systems

- **Supply chain management (SCM):**
- **Determines**:
  - What supplies are required for value chain
  - What quantities are needed to meet customer demand
  - How supplies should be processed into finished goods and services
  - How shipment of supplies and products to customers should be scheduled, monitored, and controlled

Figure 1.16

The Value Chain of a Manufacturing Company

Managing raw materials, inbound logistics, and warehouse and storage facilities is called *upstream management*. Managing finished product storage, outbound logistics, marketing and sales, and customer service is called *downstream management*.

Upstream management

Raw materials

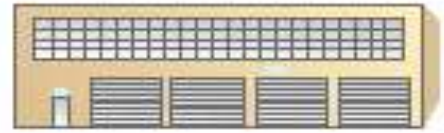


Inbound logistics



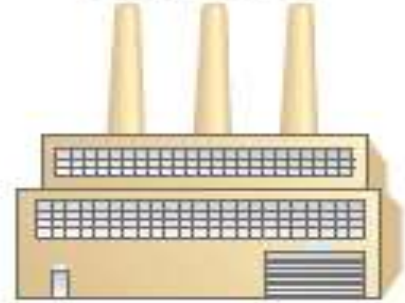
Inbound tracking systems

Warehouse and storage



Raw material inventory control systems

Production



Process control systems

Finished product storage



Automated storage and retrieval systems

Outbound logistics



Distribution planning systems

Marketing and sales



Promotion planning systems

Customer service



Customer service tracking and control systems

Downstream management

Organizations and Information Systems

- **Customer relationship management (CRM) programs:**
  - Help companies manage all aspects of customer encounters including marketing and advertising, sales, customer service after the sale, and programs to retain loyal customers
  - CRM software can be purchased as a service and delivered over the Internet or can be installed on corporate computers

Organizations and Information Systems

Innovation:

- Organizations continuously improving their operations by looking for fresh, new ideas
- Can bring cutting-edge products and services that create new revenue streams
- – Can explore new markets and business approaches
- – Some IS departments are creating separate groups to explore new, innovative ideas

Organizational Culture and Change

- **Culture:**
  - Set of major understandings and assumptions shared by a group
- **Organizational culture:**
  - Major understandings and assumptions
  - May not be formally stated or documented
- **Organizational change:**
  - How organizations plan for, implement, and handle change

User Satisfaction and Technology Acceptance

- **Technology acceptance model (TAM):**
  - Specifies the factors that can lead to better attitudes about the information system, along with higher acceptance and usage
  - **<u>Factors include:</u>**
    - Perceived usefulness
    - Ease of use
    - Quality
    - Degree to which organization supports its use

Competitive Advantage

- Significant and (ideally) long-term benefit to a company over its competition
- Can result in higher-quality products, better customer service, and lower costs
- Organization often uses its information systems to help gain a competitive advantage

Factors That Lead Firms to Seek Competitive Advantage

- **Porter's competitive forces model:**
  - Rivalry among existing competitors
  - Threat of new entrants
  - Threat of substitute products and services
  - Bargaining power of buyers
  - Bargaining power of suppliers
- The more these forces combine in any instance, the more likely firms will seek competitive advantage

Strategic Planning for Competitive Advantage

- **Strategies:**
  - Cost leadership
  - Differentiation
  - Niche strategy
  - Altering the industry structure
  - Creating new products and services
  - Improving existing product lines and service

Strategic Planning for Competitive Advantage

- **Other strategies:**
  - Growth in sales
  - First to market
  - Customizing products and services
  - Hiring the best people
  - Innovation

Performance-Based Information Systems

- **Major stages in the use of information systems:**
  - Cost reduction and productivity
  - Competitive advantage
  - Performance-based management

Return on Investment and the Value of Information Systems

- **Return on investment (ROI):**
  - One measure of IS value
  - Investigates the additional profits or benefits that are generated as a percentage of the investment in IS technology
- **Earnings growth:**

– The increase in profit that the system brings

Return on Investment and the Value of Information Systems

- **Market share and speed to market:**
  - The percentage of sales that a product or service has in relation to the total market
- **Customer awareness and satisfaction:**
  - Performance measurement is based on feedback from internal and external users
- **Total cost of ownership:**
  - The sum of all costs over the life of the information system

Risk

- Managers must consider the risks of designing, developing, and implementing systems
- Information systems can sometimes be costly failures

Careers in Information Systems

- Career opportunities in IS are expected to increase and IS budgets expected to surge
- Mobile applications will be a key factor in this surge
- IS workers need communication skills, detailed knowledge of the organization, and understanding of how information systems can help the organization achieve its goals

Roles, Functions, and Careers in IS

- **Primary responsibilities in information systems:**
  - **Operations:**
    - System operators primarily run and maintain IS equipment
  - **Systems development:**
    - Focuses on specific development projects and ongoing maintenance and review

Roles, Functions, and Careers in IS

- **Primary responsibilities in information systems:**
  - **Support:**
    - Provides user assistance in hardware and software acquisition and use, data administration, user training and assistance, and Web administration
  - **Information service units:**
    - A miniature IS department attached and directly reporting to a functional area in a large organization

Other IS Careers

- New and exciting careers have developed in IS consulting, training, sales, repair, IS customer service, and IS entrepreneurs
- Other IS career opportunities include being employed by technology companies, such as:
  - Microsoft (www.microsoft.com), Google (www.google.com), Dell (www.dell.com), and many others

Global Challenges in Information Systems

- Cultural and language challenges
- Time and distance challenges
- Infrastructure challenges
- Currency challenges
- Product and service challenges
- Technology transfer issues
- State, regional, and national laws
- Trade agreements

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Thank you for your ATTENTION

Understanding More About IS

LEC 3

INFORMATION SYSTEMS TECHNOLOGY

LEARNING GOALS

- Describe the major characteristics of the digital world.
- Discuss the use of information systems in organizations.
- Describe the features of working in the digital world.
- Describe ethics and the ethical problems posed by the digital world.

Case Study: Introducing Stan & BRJ

Stan L. lives and works in a digital world

- Digital house
- Digital paper
- Portable, digital communications everywhere

Stan owns BRJ Consulting

- Completely digital company
- No offices

Stan and BRJ have a new client – the Bead Bar

Case Study: Information at the Bead Bar

- Founded in 1998 by Meredith S.
- *Create your own* bead jewelry
  - Bead Bar provides
    - Beads, Wire, String
  - Customer designs and constructs product with Bead Bar materials and instruction
- Three divisions
  - Studios – six
  - Franchises – five
  - Bead Bar on Board
- 15 full-time employees and 20 part-time
- $1.5M annual revenues

Case Study: Information at the Bead Bar

Paper forms are inefficient for the current size of the Bead Bar

- Lost orders
- Incorrect invoicing
- Fulfillment delays

Meredith needs a solution to bring the Bead Bar into the digital world

The Nature of the Digital World

- Globalization
- The digitization of goods
- Speed
- The merger of products and services
- The role of ideas
- New markets and pricing

Globalization

- The system of interconnected economies
  - Based on capital markets
  - Characterized by the use of technology
- Presents opportunities for growth
- **Requires an understanding of:**
  - Local customs
  - Local laws
  - Regulations of foreign countries
- globalization allows businesses to operate internationally, utilizing digital technologies to reach global markets, streamline operations, and enhance collaboration across borders.

The Digitization of Goods

- Change in format of many creative products
  - MP3s
  - DVDs
  - **Software**
    - Download virus protection files automatically to protect your computer
  - **Digital documents**
    - Newspapers NYTimes
    - Electronic library databases

The Digitization of Goods

- The digitization of goods involves converting physical products into digital formats or creating digital versions of goods that can be easily distributed and consumed online.
- This includes e-books, digital music, online software, and streaming media.
- The digitization process enhances accessibility, reduces costs, and enables businesses to innovate and provide new services.

Speed

- Digital availability includes the expectation of rapid availability and delivery
  - Anytime
  - Any place
- Speed of change
- Speed refers to the rapid pace at which information, products, and services are created, processed, and delivered in today's digital economy.
- With advancements in information technology, businesses can respond quickly to market changes, customer demands, and competitive pressures, leading to shorter product development cycles and faster delivery times.

The Merger of Products and Services

- Blending of tangible products with intangible services
- The service becomes a product
- The merger of products and services refers to the integration of tangible goods with intangible services, creating hybrid offerings that enhance customer value.
- **For example**, a company might sell a smartphone (product) while also providing software updates, customer support, and cloud storage services (services). This trend reflects a shift towards providing holistic solutions that address customer needs beyond just the physical product

The Role of Ideas

- The role of ideas highlights the importance of innovation, creativity, and intellectual property in driving business success in the digital age.
- In information systems, new ideas can lead to the development of disruptive technologies, business models, and processes that challenge traditional practices and create competitive advantages.

The Role of Ideas

- New ideas at the heart of IT innovations
- Digitalization makes theft much easier
  - Web-site design
  - Digital images
  - Digital content
- Protection of intellectual property
  - Patents
  - Copyrights
    - Digital watermarks - Digimarc
  - Trademarks
  - Trade secrets

New Markets and Pricing

- New markets refer to emerging opportunities for businesses to expand their reach, often driven by globalization, digitalization, and changes in consumer behavior.
- New pricing strategies, such as subscription models or dynamic pricing, are often adopted to capitalize on these markets, allowing businesses to attract diverse customer segments and optimize revenue streams.

New Markets and Pricing

Digital markets have created new businesses

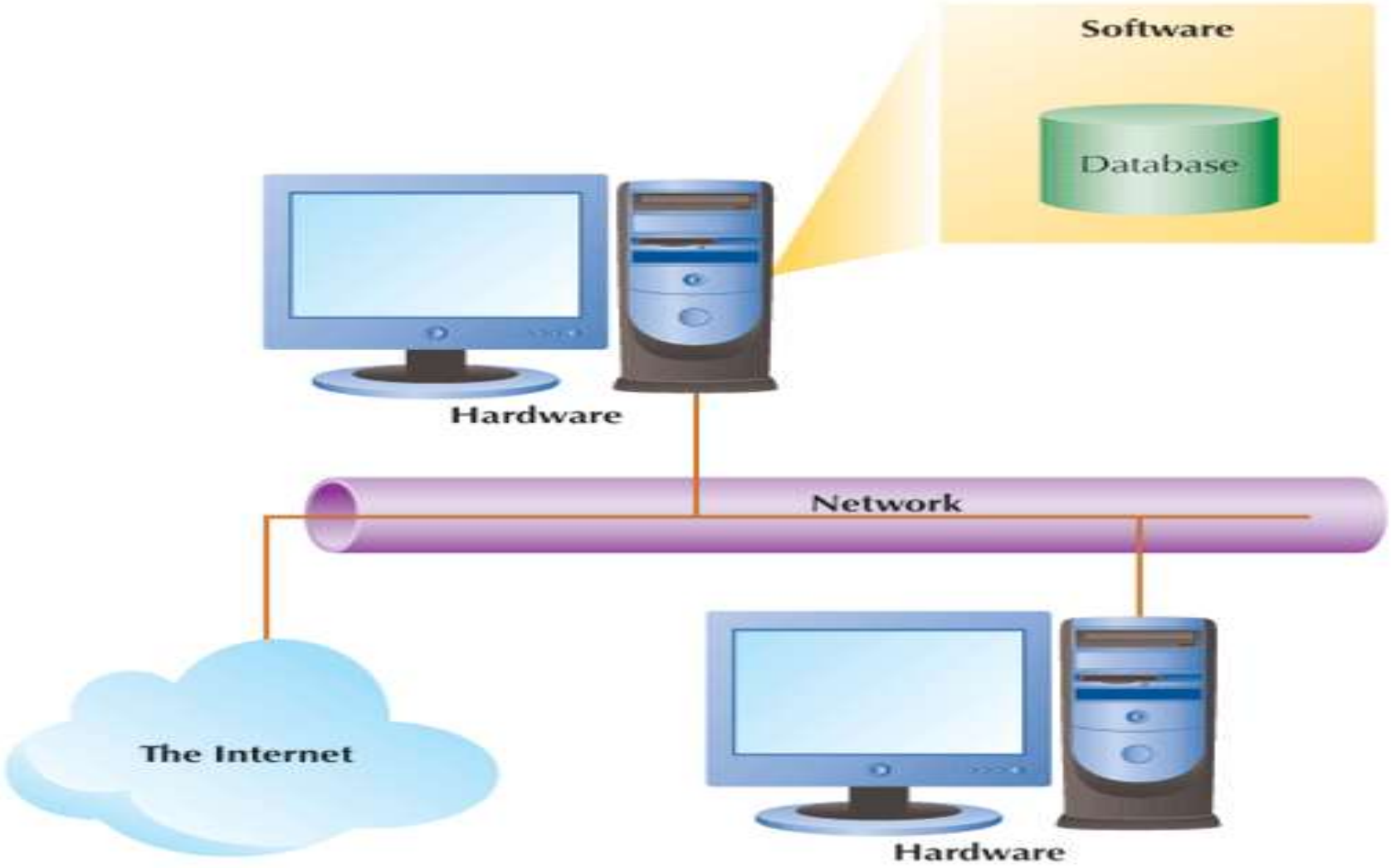
- **Travel**
  - Orbitz
  - Expedia
- **Customer-to-customer exchanges and auctions**
  - eBay
  - Yahoo auctions
- **Financial**
  - Fidelity
  - eTrade

Information Systems

- A set of interrelated information technologies that work together to process, store, collect, and distribute information

- **Major parts**
  - Hardware
  - Software
  - Databases
  - Networks

Figure 1.1 Information System Components



Information Systems in Organizations

- An organization is an administrative and functional structure applied to people working toward a specific goal.
- Understanding the organizations IT needs means understanding the **administrative** and **functional** structure.
  - Hierarchical
  - Matrix
  - Other

1. Administrative Information Systems (AIS)

- refer to the systems that support the administration and management of an organization’s operations.
- These systems play a critical role in the efficient handling of data and processes, enabling better decision-making, communication, and overall organizational performance.

1. Administrative Information Systems

- Transaction Processing Systems (TPS)
- Office Automation Systems (OAS)
- Knowledge Work Systems (KWS)
- Management Information Systems (MIS)
- Decision Support Systems (DSS)
- Executive Information Systems (EIS)
- InterOrganizational Systems (IOS)

2. Functional Information Systems (FIS)

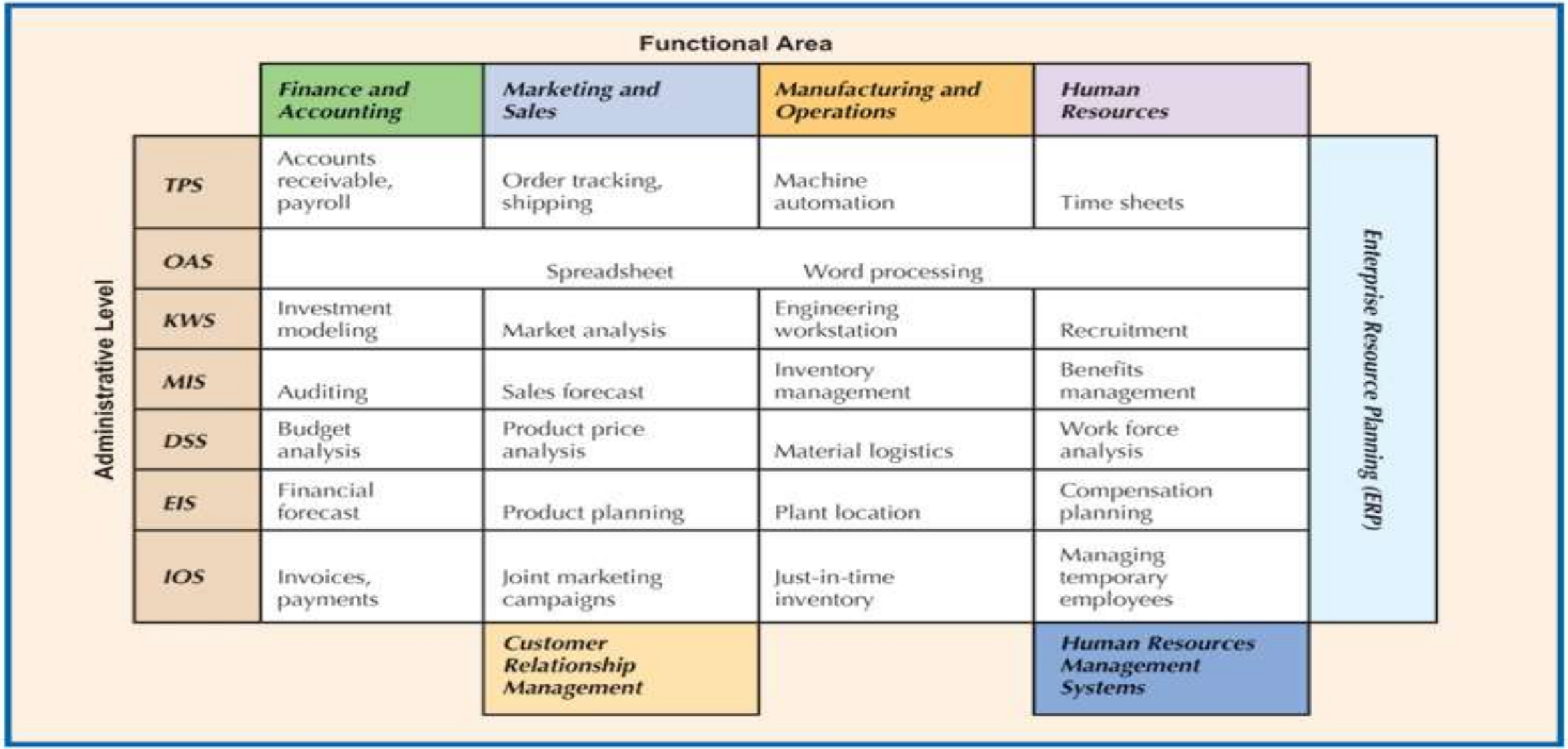
- are specialized information systems designed to support specific business functions within an organization.
- Each system focuses on enhancing the efficiency, effectiveness, and productivity of its respective area, such as finance, human resources, marketing, production, and customer service.

2. Functional Information Systems

- Finance and Accounting Systems
- Marketing and Sales Systems
- Manufacturing and Operations Systems
- Human Resources Management Systems

Interaction of Administrative and Functional IS

Figure 1.3 Administrative and Functional Information Systems



Working in the Digital World

- The digital world caused major changes in business and in the ways individuals work.
- **Creation of knowledge workers**
  - People who create new knowledge or modify existing knowledge
  - Mental not physical
  - Advanced education
    - College degree
    - Professional certifications – MCSA
- Telecommuting – working from home or other remote location

Ethics in the Digital World

- **Ethics**
  - A set of principles of right conduct
  - The rules or standards governing the conduct of a person or the members of a profession
- Ethical is not always the same as legal.
- Ethics in the digital world are confusing.
- Legal system has not kept pace with the technology developments.

Major Ethical Issues in the Digital World

- **Privacy**
  - Identify theft
  - Sale of confidential personal information
- **Intellectual Property**
  - *Old* Napster
- **Freedom of Speech**
- **Social Concerns**

Figure 1.4 Ten Commandments of Computer Ethics

Ten Commandments of Computer Ethics
1.Thou Shalt Not Use A Computer To Harm Other People.
2.Thou Shalt Not Interfere With Other People's Computer Work.
3.Thou Shalt Not Snoop Around In Other People's Computer Files.
4.Thou Shalt Not Use A Computer To Steal.
5.Thou Shalt Not Use A Computer To Bear False Witness.
6.Thou Shalt Not Copy Or Use Proprietary Software For Which You Have Not Paid.
7.Thou Shalt Not Use Other People's Computer Resources Without Authorization Or Proper Compensation.
8.Thou Shalt Not Appropriate Other People's Intellectual Output.
9.Thou Shalt Think About The Social Consequences Of The Program You Are Writing Or The System You Are Designing.
10.Thou Shalt Always Use A Computer In Ways That Insure Consideration And Respect For Your Fellow Humans.

Source: www.brook.edu/dybdocroot/its/cei/overview/The_Commandments_of_Computer_Ethics.htm

Case Study: Bead Bar Consultant

How do this chapter's issues affect the Bead Bar?

- Meredith – Bead Bar combines product and service
- Suzanne – Patents and copyrights
- Leda – Trademarks
- Mitch – Globalization
- Julia – Wants to telecommute
- Miriam – Developing a virtual organization
- Rachel – Internationalization of supply chain
- Jim – HR policies for new modes of work

Learning Goals Summary

In this chapter you have learned:

- ✓ The major characteristics of the *digital* world
- ✓ How organizations use information systems
- ✓ Some of the features of working in the digital world
- ✓ Ethics and the ethical problems of the digital world

Q????

COMPUTER HARDWARE TECHNOLOGIES

The Historical Development of Computers

- **Babbage's Analytical Engine – 1820s – 1870s**
  - Punched card program
  - Lady Ada Byron, the first programmer
- **Herman Hollerith**
  - Hollerith code
  - Punched card tabulating machines – 1890 census
- ENIAC during WWII used **vacuum tubes**
- 1960s introduced **transistor** to replace tubes
- Integrated circuits replaces transistor
- 1970s – introduction of the **microprocessor**

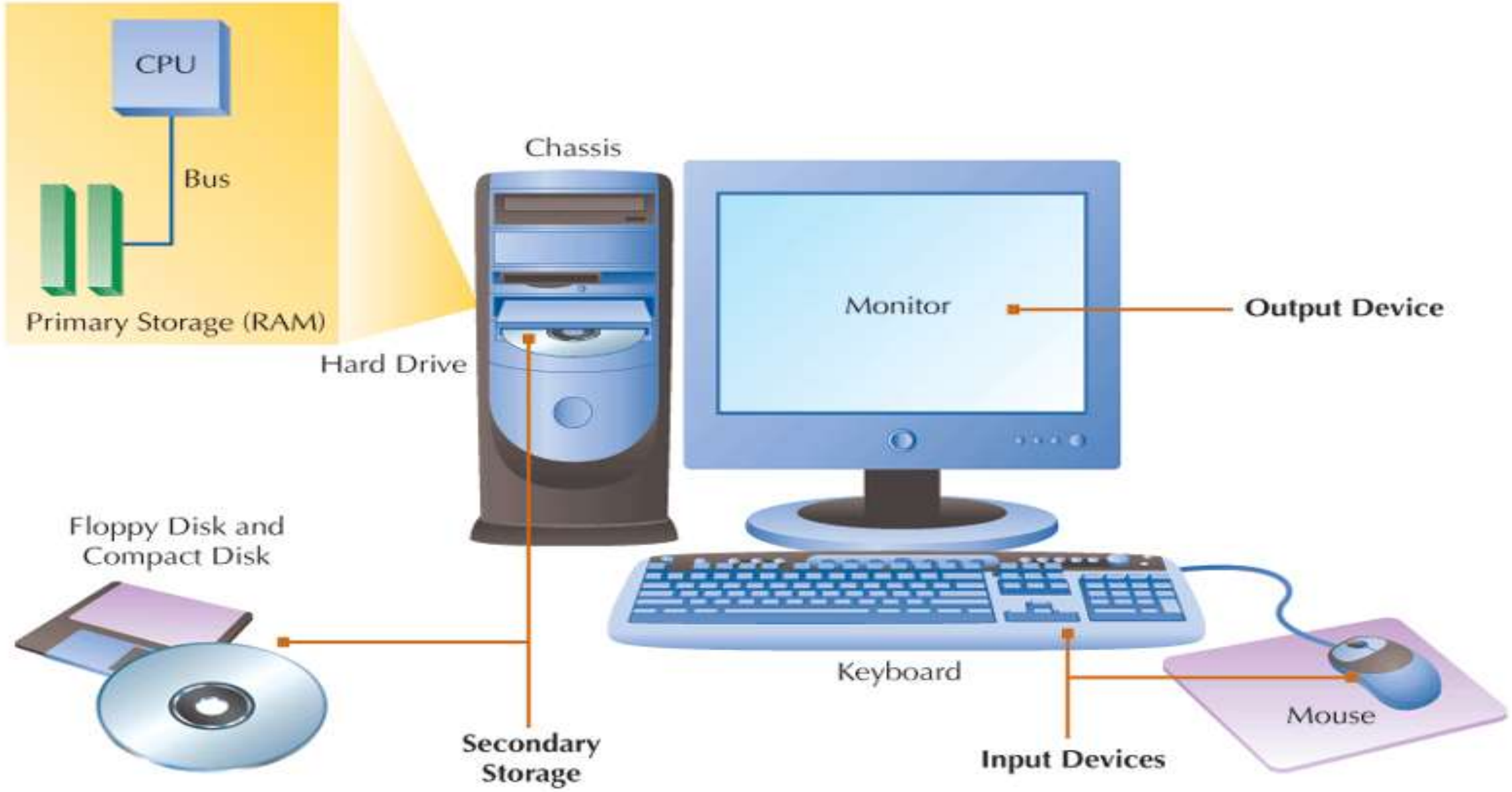
The Rise of the PC

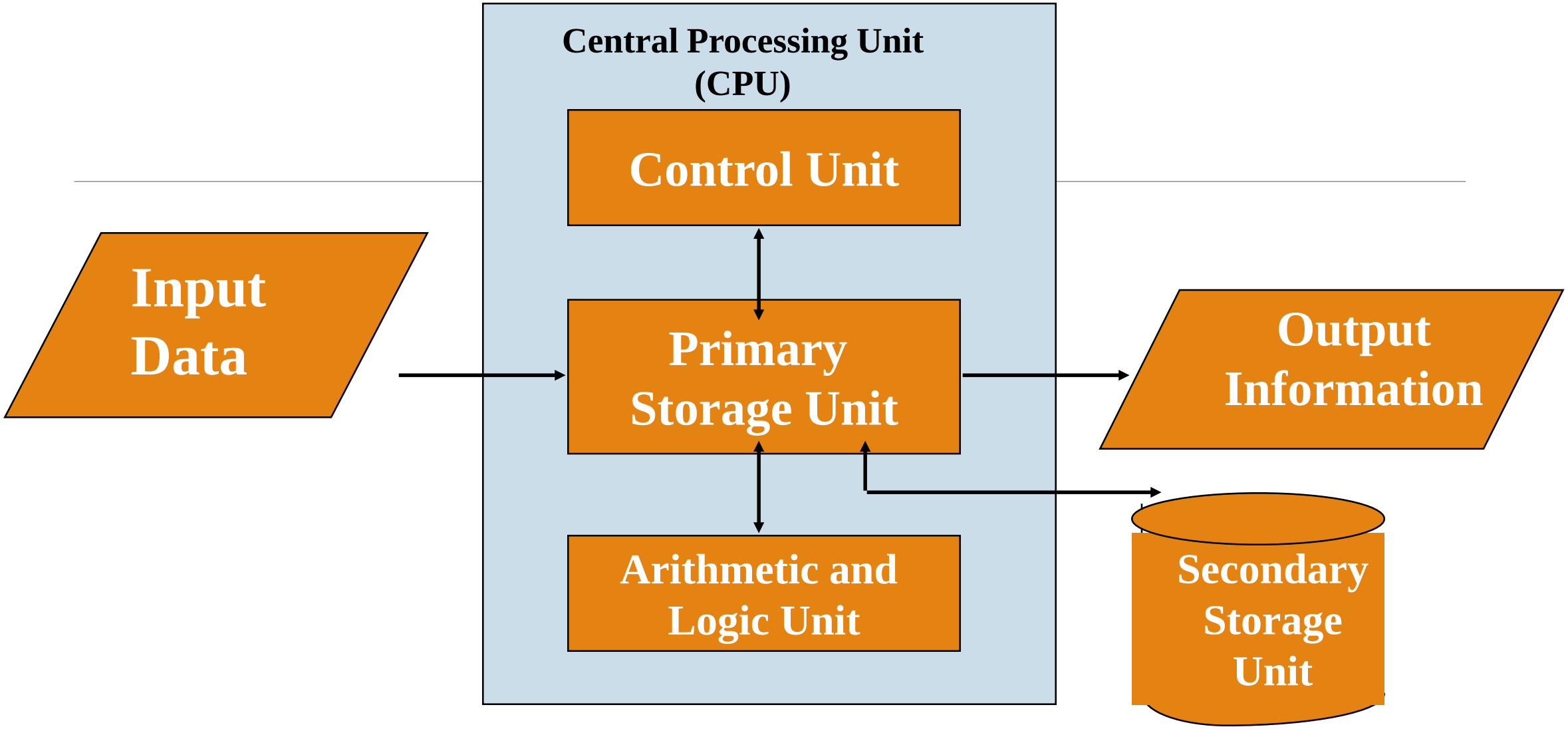
- 1974 – Altair 8800 – the first PC
- 1976 – Wosniak and Jobs – Apple I
- 1977 – Apple II
  - Plastic case
  - Keyboard
  - Capable of color graphics
- 1981 – IBM PC (introduction of MS-DOS)
- 1984 – Apple Macintosh
  - Graphical User interface (GUI) pioneered a Xerox PARC

The Core Computer Components

- Input device(s)
- Output device(s)
- Primary storage
- Secondary storage
- Central processing unit (CPU)
- Busses

Figure 2.1 Core Components of the Computer





The Computer Schematic

Input Devices

- **Human input devices**
  - Allow person to send data to the computer
  - Keyboard
  - Mouse (and other pointing devices)
- **Machine-readable input devices**
  - Send data directly to computer without human involvement
  - No human involvement means no human error
  - Usually faster than human input

Keyboard as Input Device

- Easy way to enter text, numbers, and simple commands
- Follows the layout of the typewriter
- ***QWERTY keyboard leads to RSIs***
  - Repetitive Stress Injuries
  - Requires increased attention to ergonomics of the work environment
    - Posture
    - Lighting
    - Working

Machine-readable Input Devices

- **Bar code scanners**
  - Uses light to read series of coded stripes
  - Universal Product Code (UPC) – Adams.com
  - Package tracking at UPS
- **Optical Character Recognition (OCR)**
  - Software that works with scanner
  - Translates scanned digital image to character that user can recognize and manipulate
- **Magnetic Ink Character Recognition (MICR)**
  - Character recognition technology used by banks to allow rapid routing of checks between banks

Figure 2.5 Output Devices

Output Device	Picture	Application
CRT monitor		Affordable and high-quality images
LCD monitor		Sleek styling, saves space, more expensive than a CRT monitor
Touch screen monitor		Easy input, no keyboard or mouse necessary, quick navigation through programs
Laser printer		Fast printing, high-quality work, more expensive than ink-jet printer
Ink-jet printer		Affordable color and black varieties

Output Devices

- **Monitors**
  - Cathode ray tube (CRTs)
  - Liquid crystal displays (LCDs)
  - ***Quality of display***
    - Resolution
    - Dot pitch
- **Touch screens**
  - Input and output via display device

Printers

- **Speed and resolution**
  - Pages per minute
  - Dots per inch
    - Number of ink dots to fill a square inch
    - Higher DPI the greater the page clarity
- **Impact**
  - Create image by striking paper and ribbon
  - Dot-matrix
- **Non-impact**
  - Create image by spraying or rolling ink on the page

Non-impact Printers

- **Ink-jet technology printers**
  - Spray ink on the paper
  - Small
  - Quiet
  - Color is readily and cheaply available
- **Laser printers**
  - Laser heats drum which rolls ink (toner) on paper
  - Can be faster than ink-jet
  - Color more expensive than ink-jet

Figure 2.6 Multimedia Input and Output Devices

Multimedia Device	Picture	Application
Speakers		Outputs sound, such as music
Digital camera		Takes pictures, which can be stored and edited on a computer
Camcorder		Captures video, which can be stored and edited on a computer
Scanner		Copies paper documents and pictures into a computer
TV		Displays computer output
Game pad		Controls computer games
Microphone		Enters voice commands or sound files on a computer
Joystick		Controls computer games

Multimedia I/O Devices

- Music, images, and video
  - Convert content to digital form
  - How MP3s work at HowStuffWorks.com
  - Digital cameras ; digital camera phones
  - Digital Video DVD recorders – Sony Handicam
- Game controllers
- Virtual reality (VR)
- Voice recognition
- Text-to-speech
- Brain wave input

Primary Storage

- Digital data stored in bits – BInary digiTs
- Combination of 8 bits is called a byte
- Different combinations of bits represent different symbols [**A**]
  - ACSII [10000010]
  - EBCDIC [10100010]
- All data and programs are represented in bits
  - Video
  - Pictures
  - Text

Primary Storage (Continued)

- The ***main memory*** of the computer
- Stored on semiconductor chips
  - **RAM – random access memory**
    - Volatile
    - Cache memory
      - Internal – ultra high speed and VERY expensive
      - External – very high speed and less expensive than internal cache
  - ***ROM – read only memory***
    - Non-volatile
- Primary storage holds all programs that are running and the data those programs use

Table 2.1 Data Storage Units

NAME	STORAGE AMOUNT	APPROXIMATE EQUIVALENT
Byte	String of 8 bits	1 character
Kilobyte	1,024 bytes	½ typewritten page
Megabyte	1,048,576 bytes	1 digital picture
Gigabyte	1,073,741,824 bytes	Beethoven's 5th Symphony on CD
Terabyte	1,099,511,627,776 bytes	2,000 CDs
Petabyte	1,125,899,906,842,624 bytes	160,000 DVDs (more than half of all theatrical releases)
Exabyte	1,152,921,504,606,846,976 bytes	½ the amount of information generated worldwide in a year (5 exabytes = all words ever spoken by human beings)
Zettabyte	1,180,591,620,717,411,303,424 bytes	As much data as grains of sand on all the world's beaches
Yottabyte	1,208,925,819,614,629,174,706,176 bytes	As much data as the number of atoms in 2 tablespoons of water

Sources: www.cio.com/archive/011501/tl_numbers.html
www.jamesshuggins.com/h/tek1/how_big.htm
www.magictree.com/dataprefixes.htm

Secondary Storage

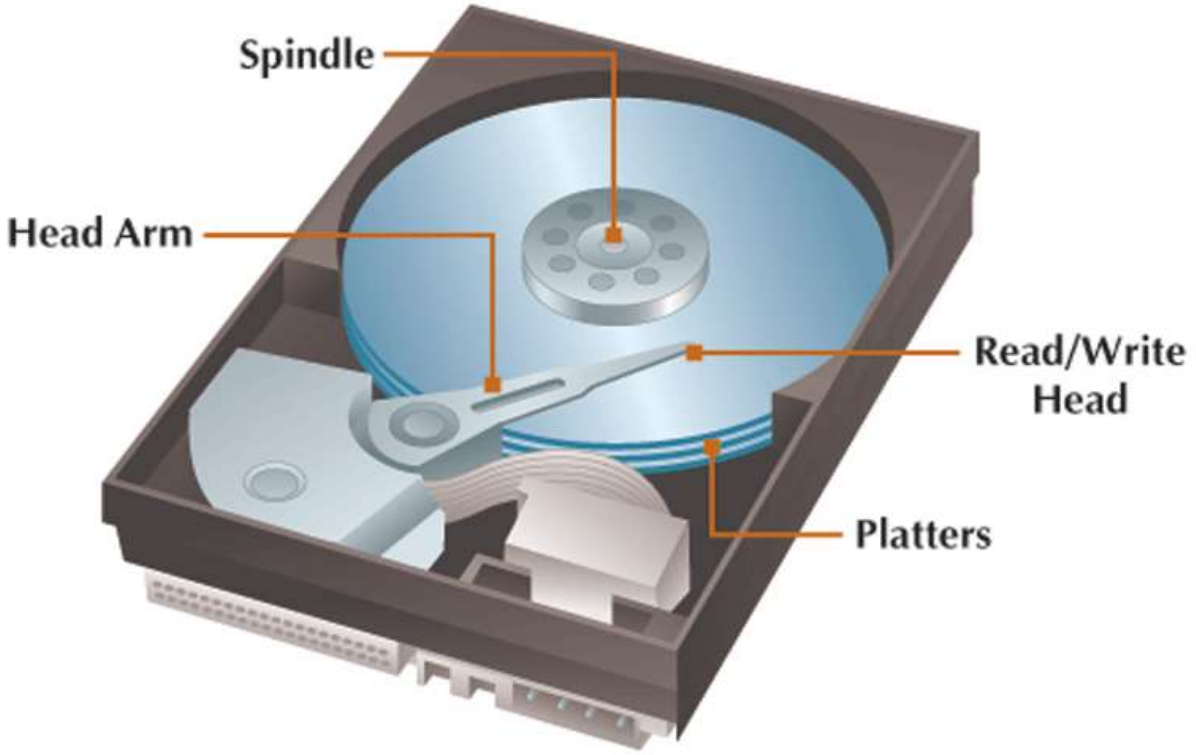
- Nonvolatile storage of digital data
- Magnetic or optical
- **Magnetic media**
  - Sequential access (tape) or direct access (HDD)
  - Hard disk drives
    - Metal disk with metallic coating
    - Disk spins at speeds of up to 7000 rpm
    - RAID – redundant array of inexpensive disks
  - **Floppy** disks
  - **Magnetic** tape
  - Zip disks and other magnetic storage alternatives

Hard Disk Drives and Direct Access



Sources: www.dis.unimelb.edu.au/mm/hwtute/peripheral_devices/tapes.htm, www.dis.unimelb.edu.au/mm/hwtute/peripheral_devices/disk.htm

Figure 2.8 A Computer Hard Disk



Source: www.4-winner.com/computers/harddriveinfo.htm

Optical Secondary Storage

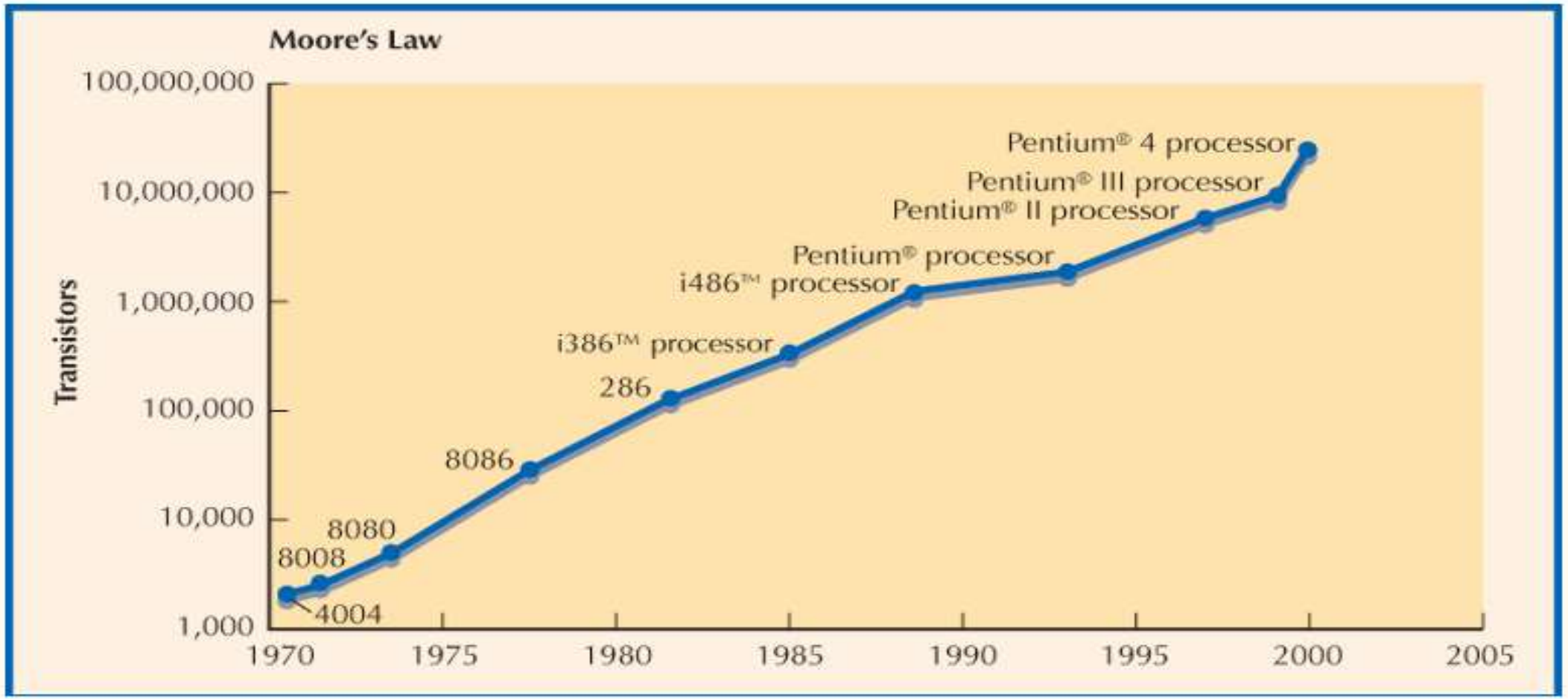
- **Compact disks (CDs)**
  - 600 MB capacity
  - CD-ROM
  - CD-R
  - CD-RW
- **Digital versatile disks (DVDs)**
  - 2 – 17 GD capacity
  - DVD
  - DVD+/-R
  - DVD+/-RW

The Central Processing Unit

- **Two main components**
  - Arithmetic and logic unit (ALU)
  - Control unit (CU)
  - Registers are used as *scrap paper* – intermediate results
- Speed measured in hertz
  - Megahertz – millions of instructions per second
  - Gigahertz – billions of instructions per second

Moore's Law on Processor Speed

Figure 2.10 Moore's Law

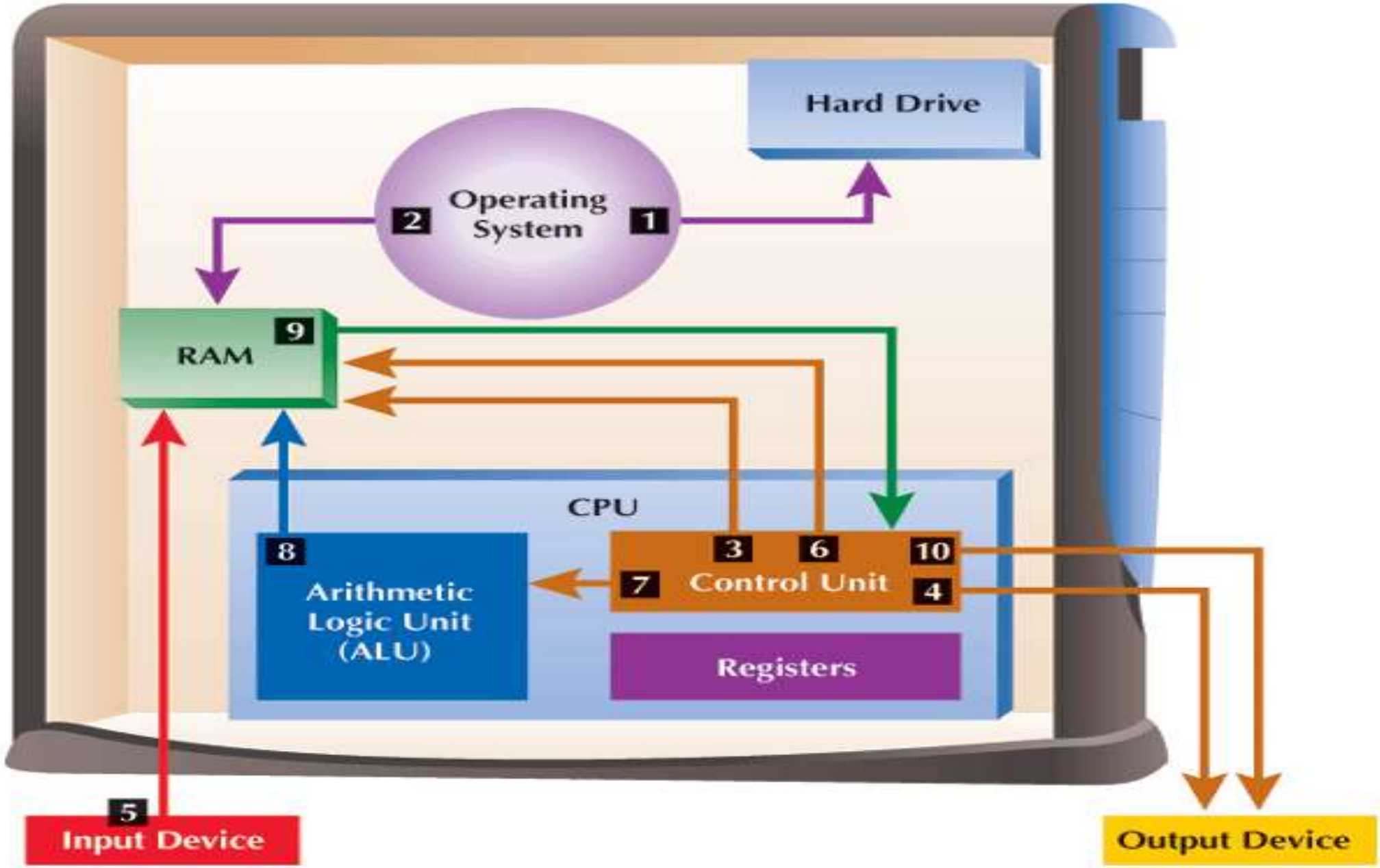


Source: ftp://download.intel.com/intel/intelis/museum/arc_collect/history_docs/pdf/mlawgraph.pdf

Computer Program Instruction Execution

- **Connect all the pieces together**
  - Bus – electrical pathway
  - All digital data moves from component to component on the bus
- **Executing an instruction**
  - Control unit **fetches** next instruction from primary storage and stores it onboard the CPU
  - The control unit **decodes** the instruction
  - The ALU **executes** the instruction possibly accessing other data in primary storage
- Cycle is repeated millions of times per second

Figure 2.12 A Program Running on a Computer



Types of Computers

- Supercomputers: the largest (***single purpose***)
- Mainframes: the first (**multiple purpose**)
- Minicomputers: the first effort to achieve reduced size (2 to 200 user at a time)
- Microcomputers:
  - Workstations
  - Desktop computers
  - Notebook computers (*laptops*)
  - Handheld computers (*PDAs*)

Learning Goals Summary

- In this chapter you have learned:
  - The major components of modern PCs
  - How the components of a computer work together
    - The role of the CPU and its operation
    - Input devices and how they operate
    - Output devices and how they operate
    - Multimedia and alternative I/O devices
    - The role of primary storage
    - Secondary storage devices and operation
  - The various types of computers

ASSIGNMENT

1. Mention another input and output devices those not mentioned in the lecture
2. Mention Other types for Keyboard
3. Mention Other types for Mouse

Q????

SOFTWARE TECHNOLOGIES

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Learning Goals

- Identify the different types of systems software.
- Explain the main functions of operating systems.
- Describe the various types of applications software and how they are used.
- Describe the software development process.
- Explain the main considerations for upgrading software and preventing piracy.

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Bead Bar Software Requirements

COMPUTER SOFTWARE REQUIREMENTS

- Meredith – Buy or hire a programmer?
- Suzanne – Software to manage employees and inventory in studios
- Leda – Easy to use
- Mitch – Contact management and presentation software
- Julia – Tax software and office suite, including spreadsheet
- Miriam – Digital picture editing
- Rachel – Software will need to be maintained and updated
- Jim – Employee benefits

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Main Types of Software

- **Systems software**
  - Programs that manage computer hardware and application software
  - **Examples**
    - Operating systems (e.g. Windows, Linux)
    - Language translators (C++, Visual Basic)
    - Utility programs (Norton Utilities)

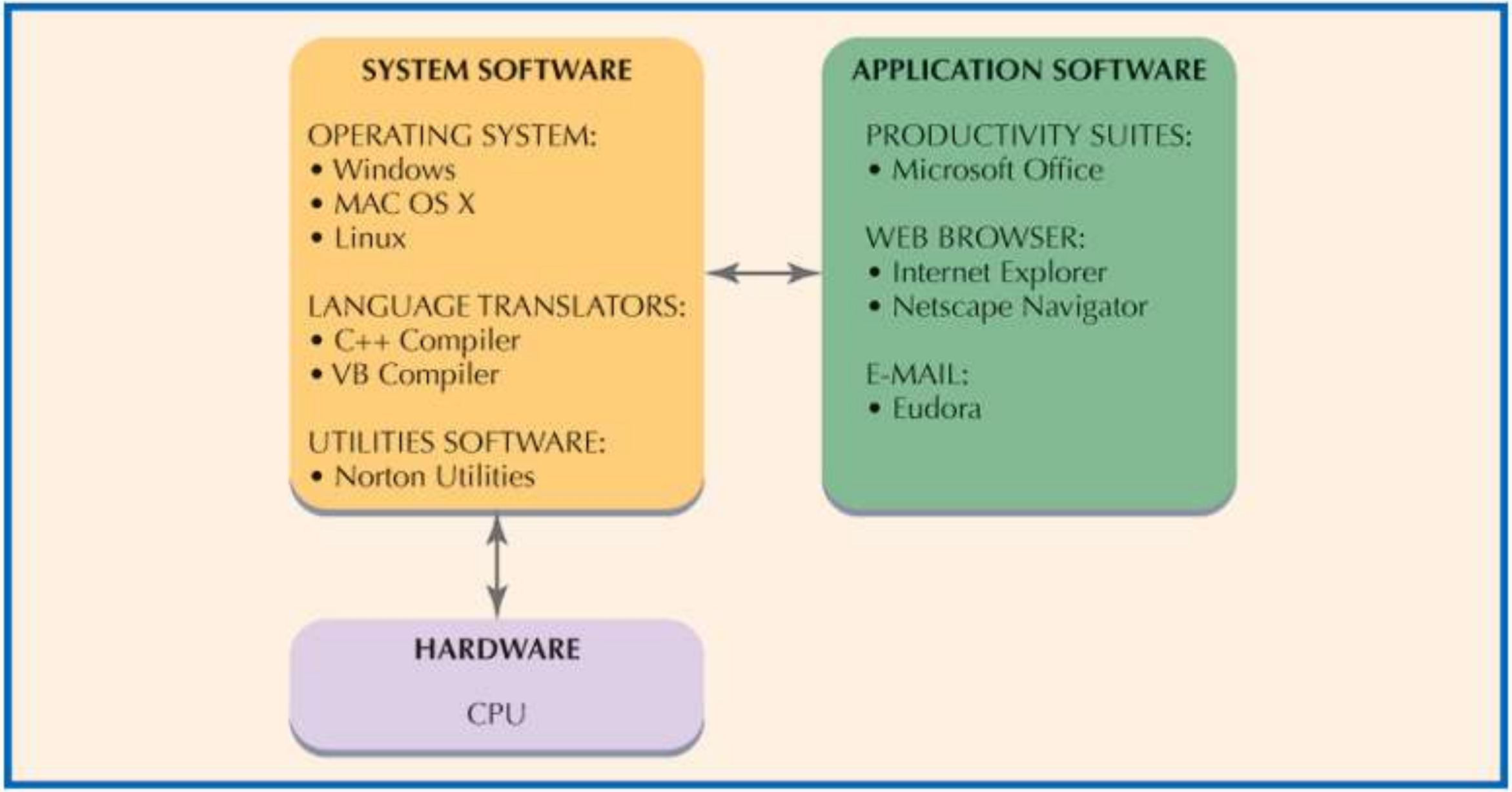
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Main Types of Software

- **Application software**
  - Programs that allow users to accomplish specific tasks
  - **Examples**
    - Word processing
    - Web browser
    - Tax preparation

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Figure 3.1 Relationship of Software to Hardware



Operating Systems (OS)

- Software platform on which other programs run
- Provides a connection between application programs and the computer hardware

Major tasks

- Starting the computer (booting or boot strap)
- Managing files
- Program and memory management
- Ensuring security
- Providing a user interface

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Managing Files

- OS keeps track where all files are located
- Creates and manages directory structure
  - Free space
  - Used space
  - Formerly used space (see defragmentation)

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Program and Memory Management

- **OS is responsible for**
  - Sending a program to the CPU
  - Allocating primary storage for the program
  - Controlling other devices the program requires
- **Multiprogramming and Multitasking**
  - More than one program in memory at once
  - OS keeps track of memory for each program
  - One program runs until it needs data from the user, then the OS switches to the other program
- **Time sharing**
- **Virtual memory**

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Figure 3.3 Multiprogramming

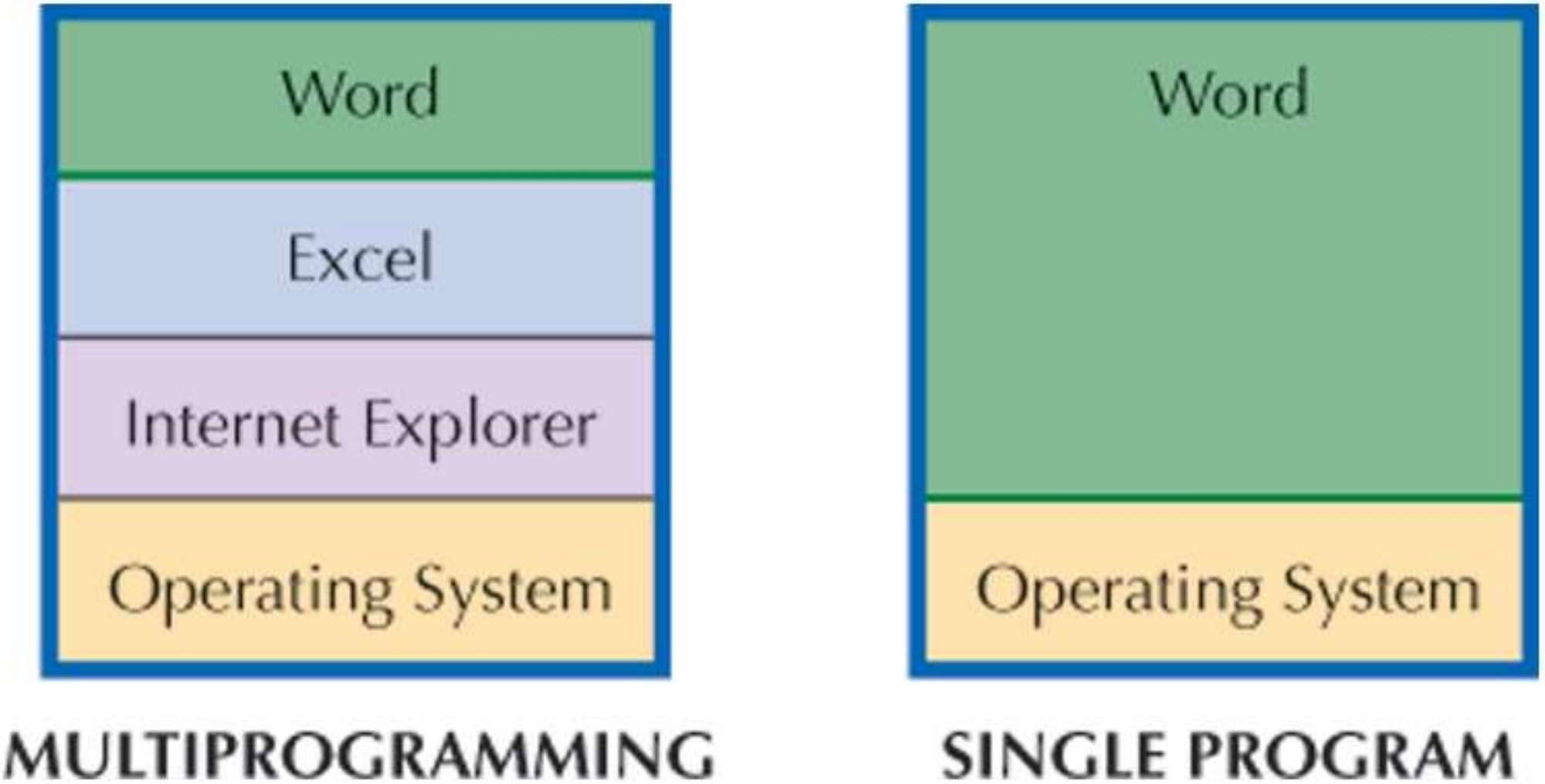
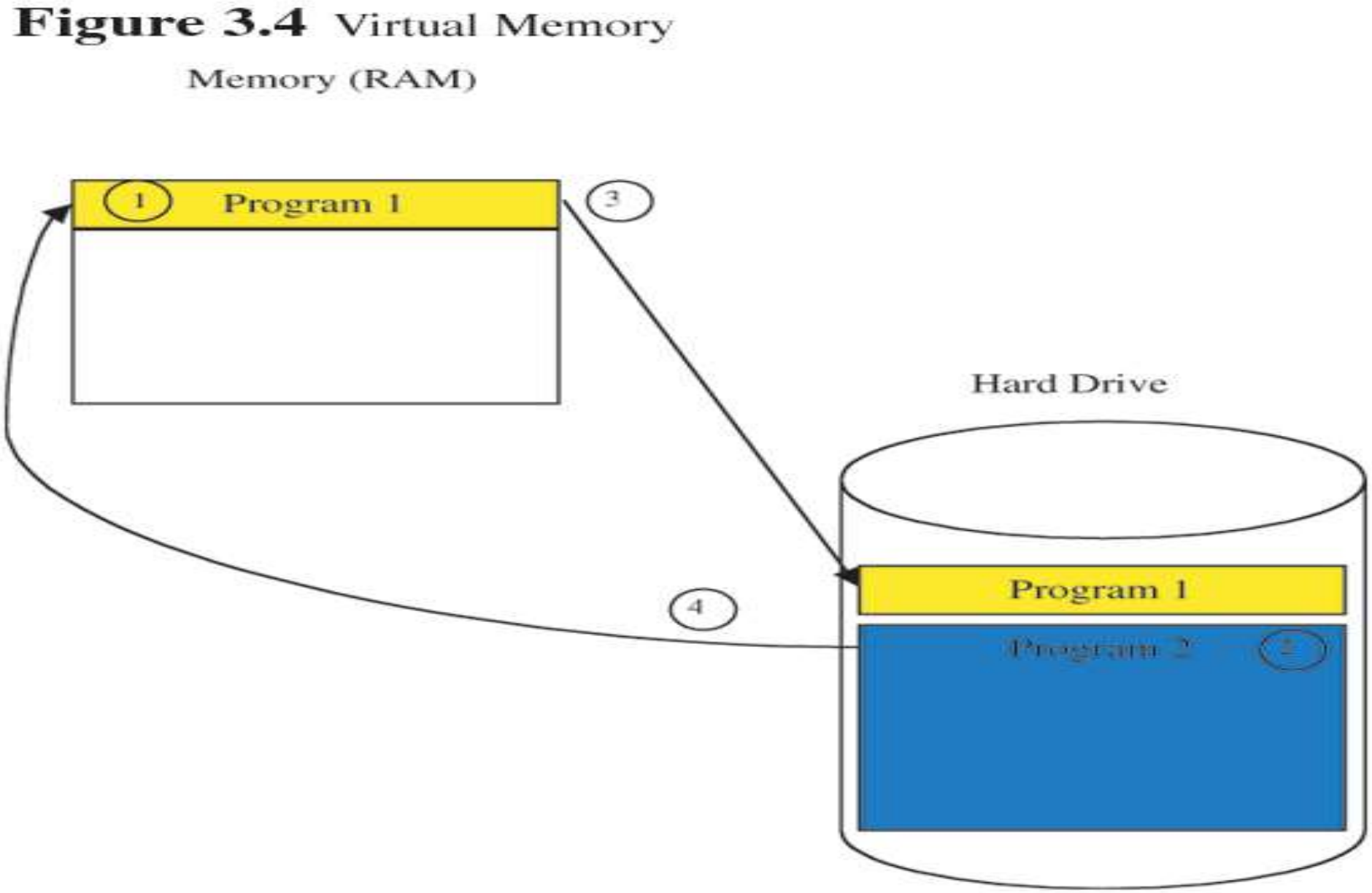


Figure 3.4 Virtual Memory



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Security

- **Access control** – Who has access to the computer; needed when network connected
- **File permissions** – Read and/or modify security permissions for individual files
- **Logging** – Audit trail of who has accessed the computer and what actions they performed

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Providing a User Interface

- Text or command line interfaces
- Graphical User Interfaces (GUIs)
- Voice interface

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TABLE 3.1 Major Operating Systems

Operating System	Release Date
	Chronological Order
UNIX	Early 1970s
DOS	1980
Windows 1.0	1983
Macintosh	1985
Windows 3.0	1990
Linux	1991
Windows NT 3.1	1993
Windows 95	1995
Macintosh 8.5	1998
Windows 98	1998
Windows Me	2000
Windows 2000	2000
Mac OS X 10.0	2001
Windows XP	2001

Choosing an OS

- **Support the applications the business needs to run!!**

Other considerations

- **Stability**
  - How often does OS *crash*?
  - How does the OS recover from crashes?
- **Security**
  - Does the OS provide the required level of security?
- **Ease of use**
  - Is the OS easy to learn and to use?

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Utility Software

- **Accomplish common tasks and maintenance jobs**
  - Defragmentation
  - Virus protection
  - Edit the Windows registry
  - Remove Web cookies from HDD

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Application Software

- **Two categories**
  - **Commercial-off-the-shelf (COTS)**
    - Productivity suites (e.g. Microsoft Office)
      - Word processing
      - Spreadsheet
      - Database
      - Presentation
      - Web browser
      - Desktop publishing
      - E-mail
    - Enterprise-wide Systems
      - ERP and CRM
    - Freeware vs. Shareware (not *free*)
  - **Customized**
    - Developed by programmers to solve a specific problem for a business

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Developing Custom Software

- Step 1 – **Plan!** – determine the requirements
- Step 2 – **Design** – develop a design for the software
- Step 3 – **Write** – Use a programming language (which one?) to write the software
- Step 4 – **Test** – Ensure the program works correctly and satisfies the requirements
  - Errors found during the testing step may require a repeat of any of the previous steps

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Programming Language Generations

Table 3.3 Programming Language Generations

	First Generation (Machine Language)	Second Generation (Assembly Language)	Third Generation	Fourth Generation
Example	1100110111	Mov bx, 37h	Int variable, variable2	SELECT * from Student
Program used	None	Assembly compiler	Pascal, C, C++, Visual Basic	SQL
Description	Computer understands only ones and zeroes	Uses simple words and numbers as memory devices	English-like syntax uses a compiler to convert to ones and zeroes	Easier for nonprogrammers to use

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Markup Languages

- Not really a programming language
- Uses markup codes, or *tags*, to determine how a document is to be formatted
- Hypertext Markup Language (HTML)
  - Markup tags that determine how pages for the World Wide Web are to be displayed
  - Not machine specific
  - Same codes used for Windows PC as for an Apple PC

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HTML

Figure 3.7 Hypertext Markup Language

Information Systems TECHNOLOGY

Software

Hardware

Programming Languages

The HTML behind the Webpage above

```
<html>

<head>
<title>New Page 1</title>
</head>

<body>

<p>Information Systems TECHNOLOGY</p>
<p><b>Software</b></p>
<p><font color="#FF0000">Hardware</font></p>
<p><i><font color="#0000FF">Programming Languages</font></i></p>

</body>

</html>
```

Object-Oriented Programming (OOP)

- Developed to provide reusability of components

Object *class*

- Objects with similar characteristics
- Easy to create new instances of the object

Inheritance

- Create new object classes (e.g. *sub*classes)
- Ability of a subclass to inherit all the characteristics and processes of the parent class

OOP makes programmers more productive

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Software Upgrades

- Even *free* upgrades can be very expensive
  - Cost for installation
  - Cost for any retraining
  - Disrupt normal operations
  - Compatibility issues
- New software will need compelling new features to justify the *cost*

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Software Piracy

- Making a copy of software without paying for it
- Ownership versus license to use
- Companies responsible for actions of employees who install or download software onto company machines

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Bead Bar Software Issues

HOW SOFTWARE ISSUES AFFECT THE BEAD BAR

- Meredith – No software piracy permitted
- Suzanne – Office suite should include database; employee scheduling and tracking of hours worked
- Leda – Can we find COTS franchise management software?
- Mitch – GUI required
- Julia – QuickBooks required
- Miriam – Desktop publishing tools needed
- Rachel – Bar code scanner software for inventory tracking
- Jim – How are computers and software going to change employee jobs, training, and recruitment?

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Learning Goals Summary

In this chapter you have learned:

- ✓ The different types of systems software
- ✓ The main functions of operating systems
- ✓ How to describe the various types of applications software and how they are used
- ✓ The software development process
- ✓ The main considerations for upgrading software and preventing piracy

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Q???

DATABASE TECHNOLOGIES

Basic Concepts in Data Management

Field

- Individual piece of data
- Made up of one or more bytes, or characters
- Examples: name, address, phone number

Record

- Fields that are grouped together for a specific purpose

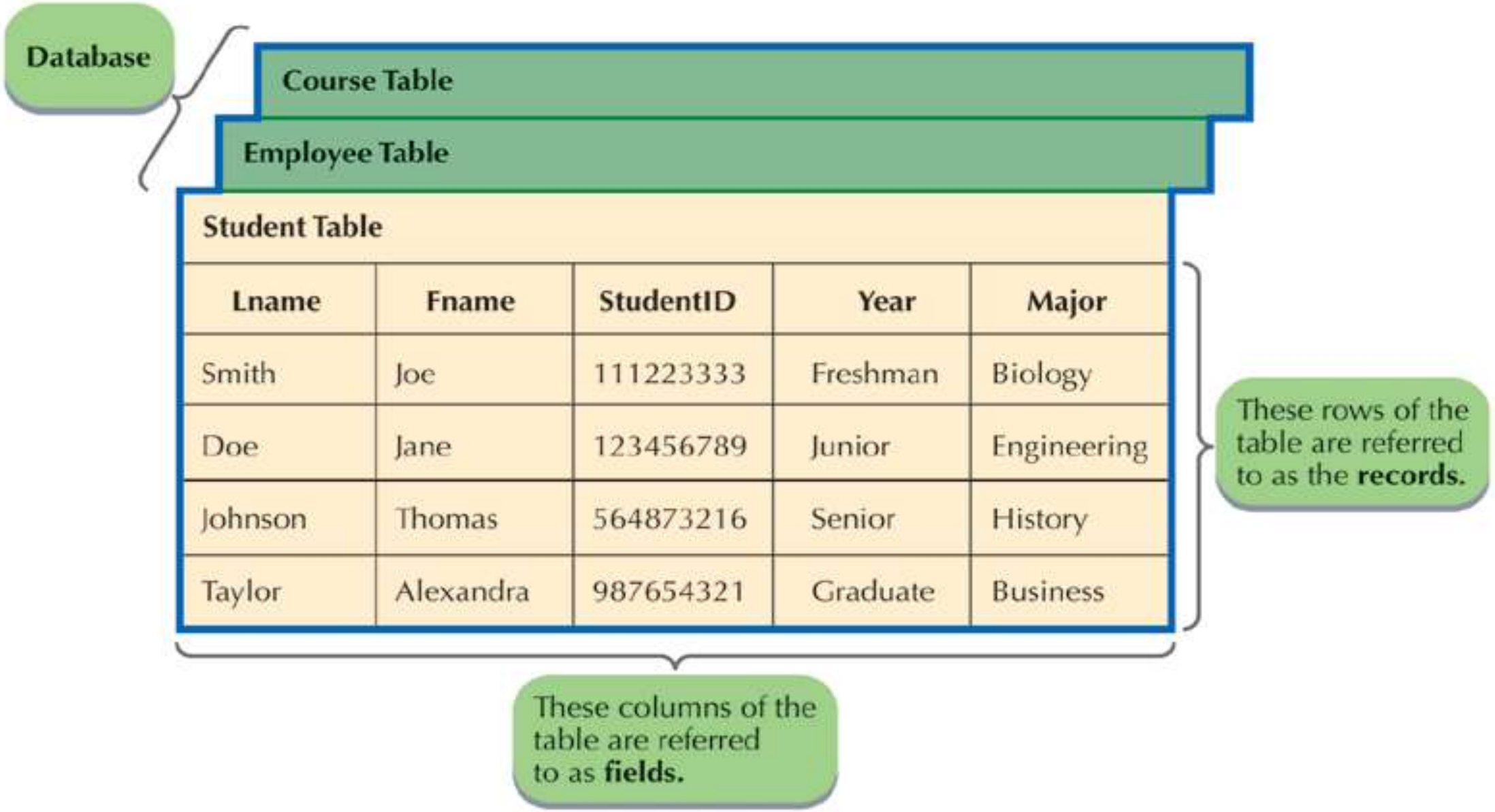
Primary key

- A field, or group of fields, that uniquely identifies an individual record
- Student id number for records describing students

More Basic Concepts

- Businesses group paper forms into a file
- Database systems equivalent of this is called a ***table***
- Files of paper forms are stored in a file cabinet
- Computer equivalent of the file cabinet is a database

Figure 4.1 Database With Multiple Tables



File Systems

- One of the simplest ways to store data
- Stores together groups of records together used by a particular software application
- **Simple but with a cost**
  - Inability to share data
  - Inadequate security
  - Difficulties in maintenance and expansion
  - Allows data duplication (e.g. redundancy)

File System Anomalies

- **Insertion anomaly**
  - Data needs to be entered **more than once** when the data is located in multiple file systems
- **Modification anomaly**
  - Redundant data in separate file systems becomes **inconsistent**
- **Deletion anomaly**
  - Failure to simultaneously delete all copies of redundant data
- **Anomalies are BAD!**

Database Defined

- **A set of logically related data** stored in a shared repository
- **Software** that creates and manipulates data is a **database management system (DBMS)**

DBMS Functions

- **Manage stored data**
- **Transform data into information**
  - Transforms the **way data** is **physically** stored into whatever **logical view** of the data that the **user chooses**
  - Hides the physical details of how the data is actually stored
- **Provide security**
  - DBMSs control who can add, view, change, or delete data in the database

More DBMS Functions

- **Allow multiuser access**
  - Controls concurrency of access to data
  - **Prevents** one user from **accessing data** that **has not been completely updated**
    - When selling tickets online, Ticketmaster allows you to hold a ticket for only 2 minutes to make your purchase decision, then the ticket is released to sell to someone else – that is **concurrency control**

More DBMS Functions (Continued)

Programming and Query Language Ability

- Data **Definition** Language (DDL) to **define** and **modify** the **structure of the data** (physical and logical views)
- Data **Manipulation** Language (DML) to allow the users to enter, modify, delete, and retrieve data from the database

Provide a Data Dictionary

- Metadata – data about data
- Data dictionary contains metadata – **data about the characteristics of databases controlled by the DBMS**

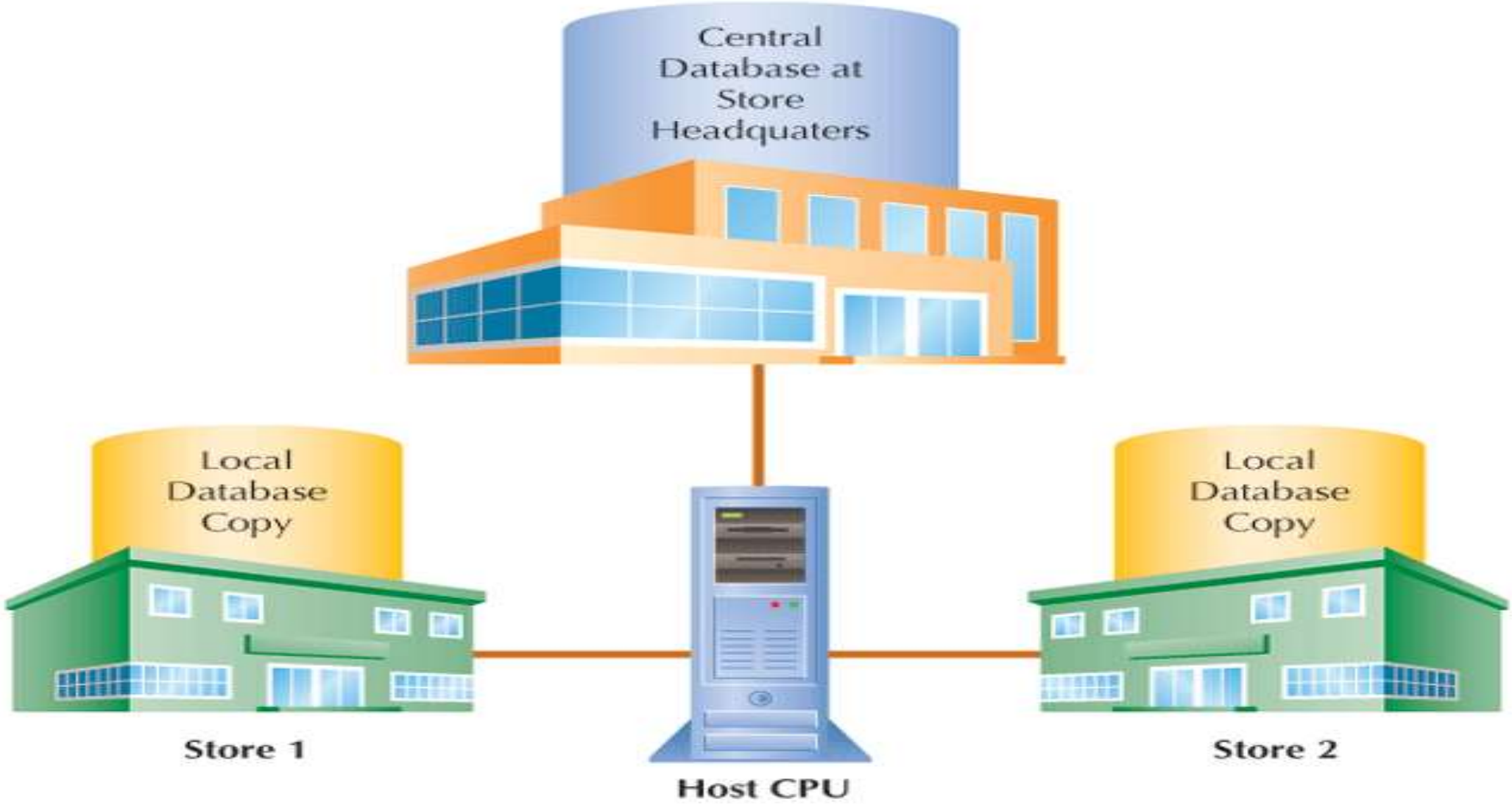
Types of DBMSs

- **Desktop**
  - Use by individuals or small groups
  - Requires little or no formal training
  - Does not have all the capabilities of larger DBMSs

Types of DBMSs (Continued)

- **Enterprise**
  - Serve multiple locations and store large amounts of data
  - Either centralized or distributed
    - **Centralized – all data on one server**
      - Easy to maintain
      - Prone to run slowly when many simultaneous users
      - No access if the one server goes down
    - **Distributed – each location has part of the database**
      - Very complex database administration
      - Usually faster than centralized
      - If one server crashes, others can still continue to operate.

Figure 4.3 A Distributed Database System



Database Models

- Database model – a representation of the relationship between structures in a database

Four common database models

- Flat file model
- Hierarchical, or tree structure, model
- Network model
- **Relational** model (this one is the most common)

Flat File Database: Address Book

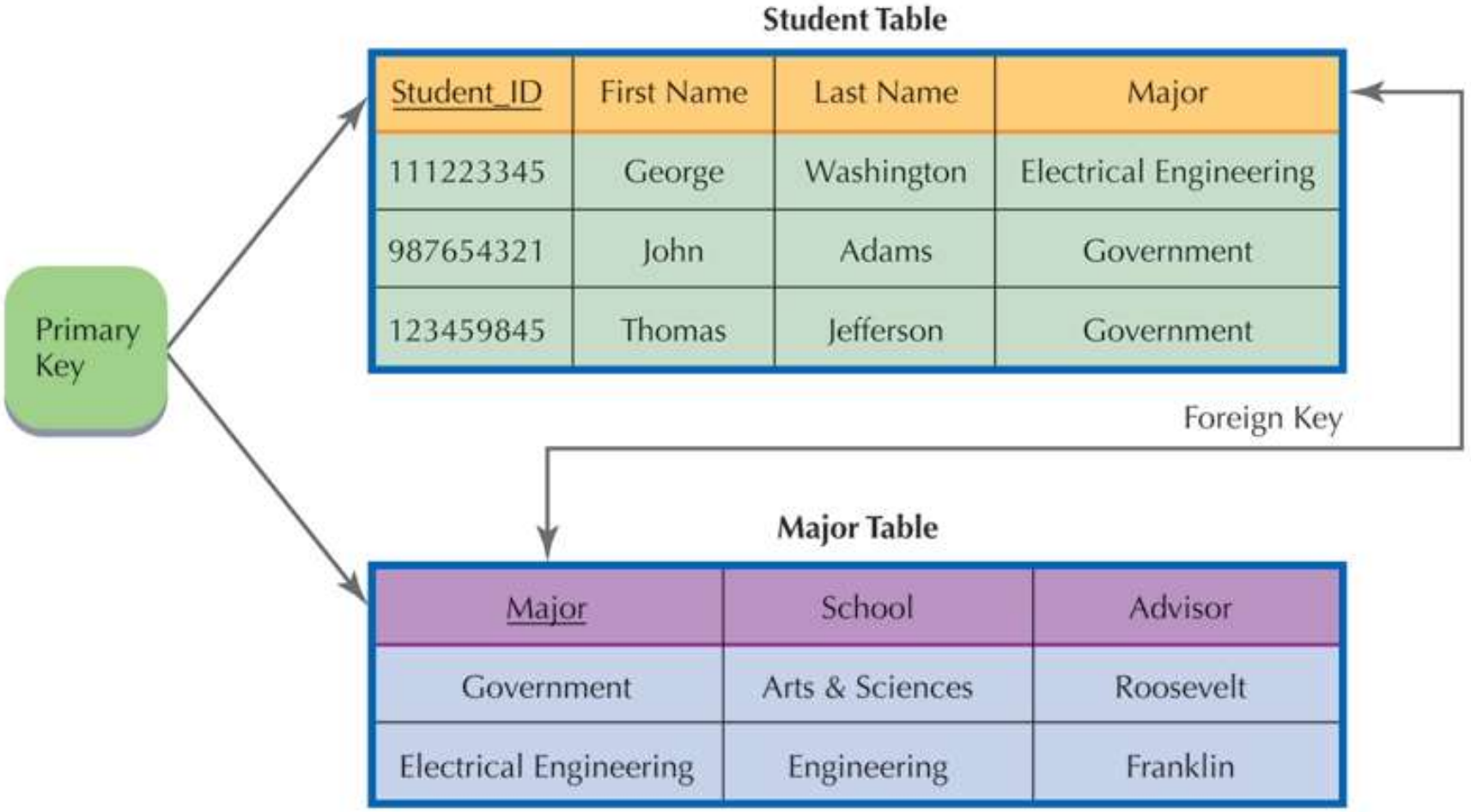
Figure 4.4 Flat File Database: Address Book

Last Name	First Name	Address 1	Address 2	City	State	Zip	Home Phone	Mobile Phone
Allen	David	123 River Road		Bowie	MD	20716	301-555-1212	
Boyle	Mary	5436 Alley Way		Greenbelt	MD	20770	301-555-9876	301-555-7887
Murray	Rita	3210 Quiet Drive	#205	Rockville	MD	20852	301-555-6677	301-555-8565
Parks	Claire	3021 Bally's Court		Annapolis	MD	21104	410-555-4132	
Smith	Gerry	87663 Colorado Ave.		Calvert	MD	23541	410-555-6971	410-555-3070
Young	Monica	6547 Boteler Land	#234	College Park	MD	20740	301-555-8216	

Relational Model

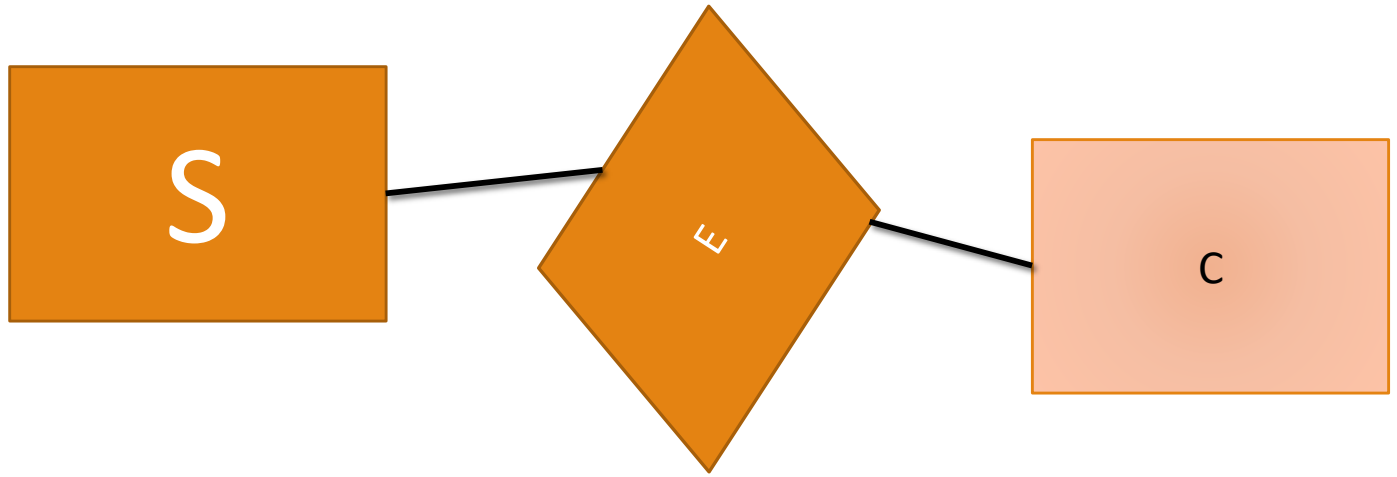
- Flexible and relatively simple to use
- Somewhat slower than hierarchical and network DBMSs
- Uses controlled redundancy to create fields that provide linkage relationships between tables in the database
  - These fields are called **foreign keys** – the secret to a relational database
  - A foreign key is a field, or group of fields, in one table that is the primary key of another table

Figure 4.7 Relational Model Example



SID PK	SN	COURSE
1	A	3
2	A	3
3	A	2

CID PK	CN	CTEACCHER
1	IS	
2	M	
3	DB	



SQL

- **Structured Query Language (SQL)**
  - Standard **DDL** and **DML** for a relational database
  - Used for
    - Creating tables (**DDL**)
    - Deleting tables (**DDL**)
    - Add, change, delete, and retrieve data (DML)
  - Although there is an **ANSI** standard specification for SQL, most vendors provide their own variety

Database Development Process

1) **Analysis** – Develop a clear **understanding** of how the organization works and what data is used

Figure 4.8 Student Registration System

Student Table

Student_ID	First_Name	Last_Name	Address	Phone
111223345	George	Washington	1 Cherry Tree Lane	333-555-2345
987654321	John	Adams	5 Boston St.	444-555-4356
123459845	Thomas	Jefferson	2 Virginia Way	888-555-8735

Course Table

Course_Num	Title	Credits
MIS 101	Introduction to Management Information Systems	3
Econ 203	Macroeconomics	3
Fin 105	Introduction to Finance	3

Course_Offering Table

Course_Num	Section	Semester	Year	Student_ID	Grade
MIS 101	2	Spring	2004	111111	A
MIS 101	1	Fall	2003	111111	F
Econ 203	1	Spring	2004	222222	B
Econ 203	1	Spring	2004	333333	A–

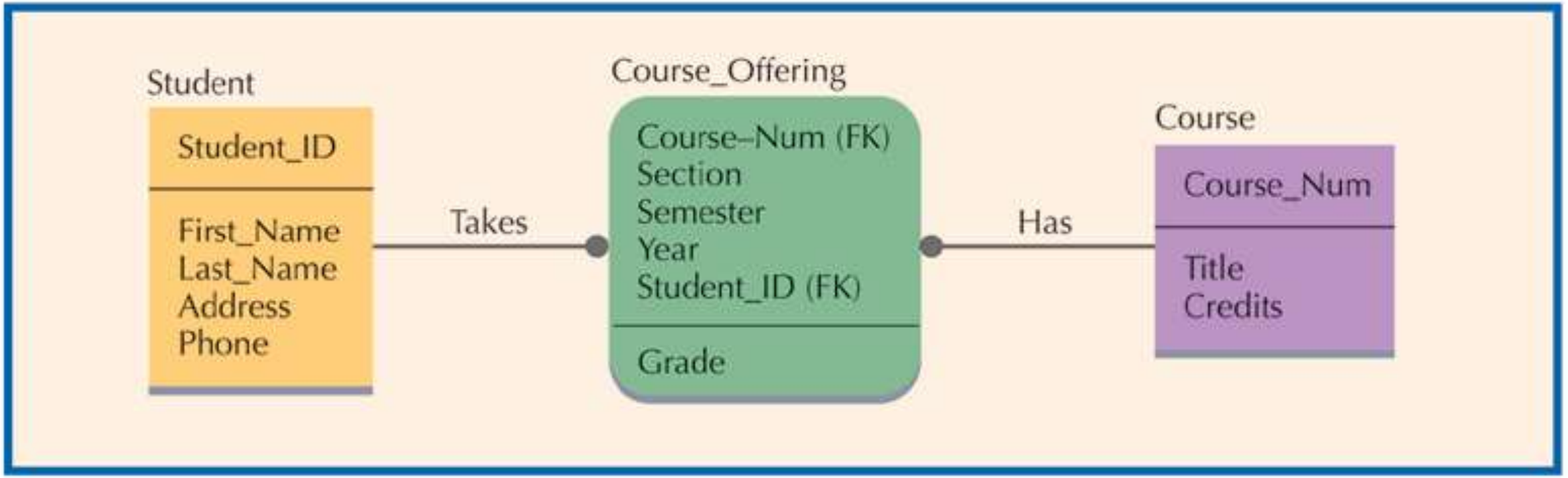
Database Development Process (Continued)

2) **Develop a conceptual model –**

- Show how data are grouped together and related to each other
- **Entity-Relationship diagrams (ERDs)** are used to record the conceptual model
- Less expensive to correct an ERD than to redesign an already constructed database

E-R Diagram Example

Figure 4.9 Entity-Relationship Diagram for a Course Registration System



Database Development Process (Continued)

3) **Develop a physical model –**

- Physical model provides specific details about each table and field in the database
- Normalization used **to remove redundant data and therefore minimize any anomalies**
- Optimize the database for performance

Database Development Process (Continued)

4) Database implementation

- Install the DBMS software
- Build the database
- Test

Database Development Process (Continued)

5) Database administration

- Ensures database efficiency
- Manages backup and restoration
- Sets up user accounts and security

Databases for Decision Making

- **Data warehouse (Collection of DBs)**
  - **Database that is**
    - **Subject-oriented** – data organized around subjects
    - **Integrated** – contains ALL data about the subject
    - **Time-variant** – data contains a time component
      - Transactional databases are accurate at a given time
      - Data warehouse contains the same data over multiple time periods e.g. a student data warehouse would contain data on what students were registered in which classes for every term covered by the data warehouse
    - **Nonvolatile**
      - The data is not updated, changed, or deleted
    - Optimized for querying and reporting
    - **NOT a transactional database**

Data Mining (تنقيب البيانات)

- Process of applying **analytical** and **statistical methods** to data to find **patterns**
- Retailers use data mining to determine purchasing patterns
- Pro football teams use data mining to scout the opposition

Advanced Database Models

- **Object-Oriented Data Model (OODM)**
  - Object class has relationships defined as well as attributes
  - OODM provides inheritance to subclasses just as in OOP
- **Hypermedia Databases**
  - Any item (called a node) linked to any other item
  - No pre-specified relationships between nodes
  - WWW is an example of a hypermedia database

Learning Goals Summary

In this chapter you have learned:

- ✓How organizations use data and information
- ✓The basic concepts of data management
- ✓Characteristics of file systems and their problems
- ✓The definitions of DBMSs and DBMS functions
- ✓ How the relational database model works
- ✓ How databases are developed
- ✓ How organizations can use data warehousing and data mining for decision making
- ✓The advanced database models and when their use is appropriate

Internet and World Wide Web Technologies

LEARNING GOALS

- Describe the Internet's current architecture.
- Identify and describe the major Internet applications.
- Define the World Wide Web and its structure.
- Understand how Internet and World Wide Web searches work.
- Describe how the Internet is governed.
- Define intranets and extranets and explain how companies use them.

Internet and WWW at the Bead Bar

- Leased lines to connect Bead Bar Headquarters with each studio would cost more than $1,000 per month.
- Meredith believes that the Bead Bar should have a Web page. Some of the uses for the web site might be:
  - Sell jewelry
  - Promotion of studios
  - Communicate with franchises
  - Help establish brand name
- Suzanne wants high-speed connections to studios
- Mitch wants to be able to search the WWW for information on the cruise industry.
- Julia is concerned about the controlling the costs
- Rachel wants to process franchise purchase requests using an intranet to improve response time.

Internet Architecture

- A Brief History of the Internet – written by some of the people who were there at the start
- **Hobbes' Internet Timeline v7.0**
- **Today's Internet**
  - Thousands of networks
  - Connected by legal agreements and commercial contracts
  - Uses TCP/IP protocol
  - **Internet service providers (ISPs)**
    - Provide most **individual users** with access to the Internet
    - Dialup connections
    - Modems and conventional phone lines
    - *x*DSL and cable modems provide broadband access

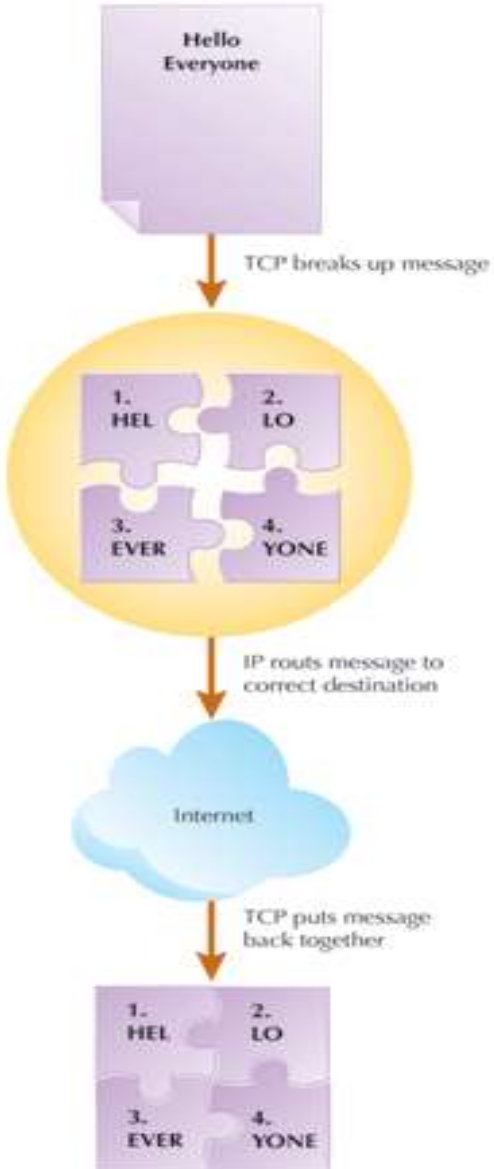
Accessing the Internet Backbone

- Much of the Internet traffic occurs among multiple ISPs
- **Network Access Points (NAPs)**
  - High-speed routers route traffic between ISPs and the Internet backbone
  - Backbone – fiber optic connections owned and operated by large telecommunications companies
    - UUNet's North America Internet network

TCP/IP

- **Communications protocol suite**
  - **Packet switched protocol**
    - No end-to-end connection is required
    - **Each message broken down into small pieces called packets**
    - Packets possibly routed to destination over different paths
  - **Transmission Control Protocol (TCP)**
    - Breaks messages into packets
    - Numbers packets in order
    - Reorders packets at the destination
  - **Internet Protocol (IP)**
    - Routes packets to the proper destination

Figure 6.3 Transmission Control Protocol (TCP)/Internet Protocol (IP)

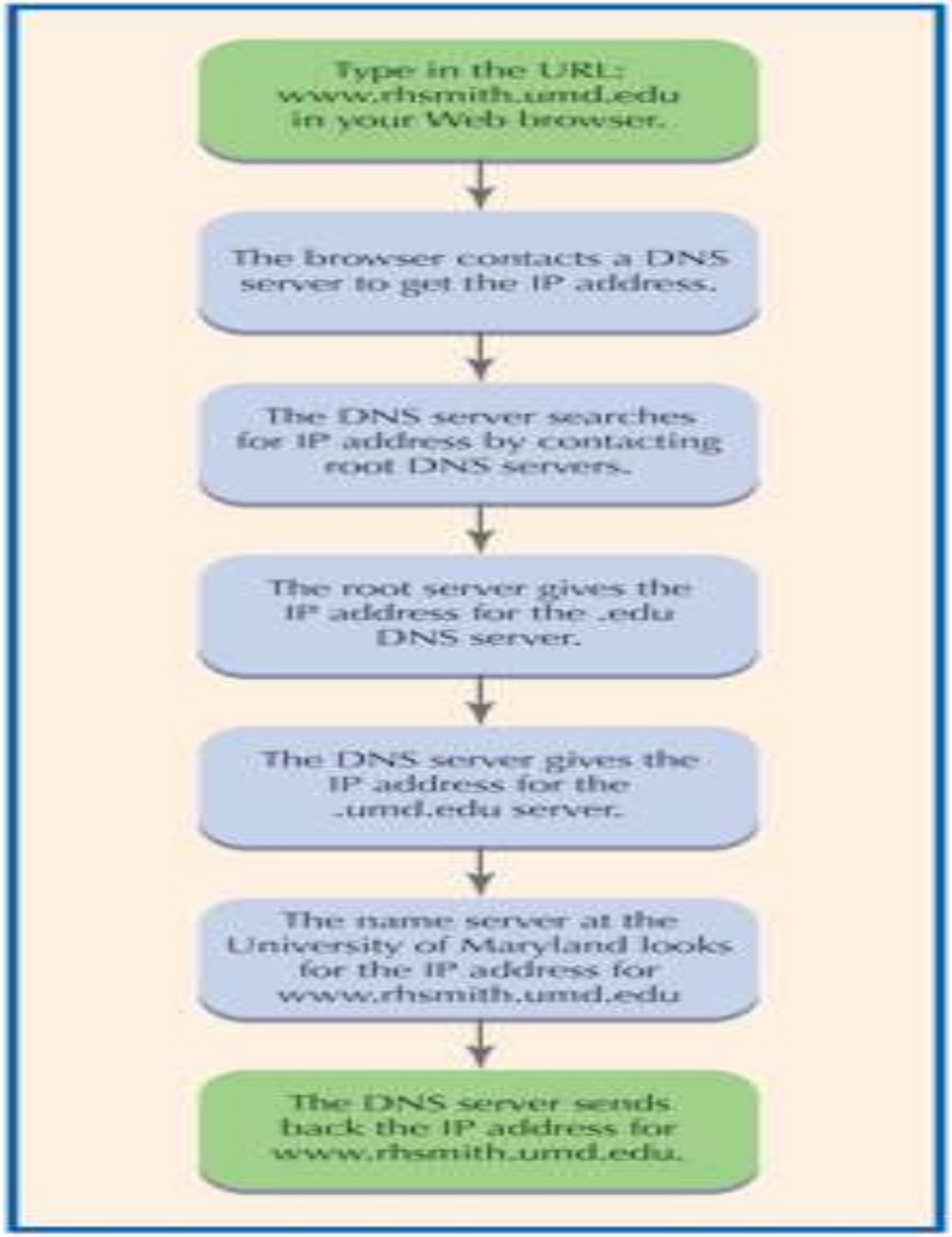


Domain Names

- Every computer connected to the Internet must have a unique IP address
  - IP address format is xxx.xxx.xxx.xxx where xxx is a number between 0 and 255
- How do we know that 207.46.245.222 is Microsoft?
- **Domain Name System (DNS)**
  - A database of Internet names
  - DNS Servers convert Internet names to IP addresses
  - Top level domains

DNS	
NAME	IP
UGC.EDU	1123098654

Figure 6.4
The Domain Name System



Internet Applications

- E-Mail
- File transfers
- Instant messaging (IM)
- Newsgroups
- Streaming audio and video
- Internet telephony
- World Wide Web (WWW)

E-Mail

- Most popular and widely used Internet application
- **30 *billion* e-mails sent every day**
  - Spam – junk e-mail messages
  - Spam costs corporate America $9 *billion* per year
- Every e-mail message contains head that describes source and destination for the message
- E-mail messages are text, but may have attachments of many types of digital data
  - Viruses often transmitted via e-mail

File Transfers

- **File transfer protocol (FTP)**
  - Protocol providing for transmission of a file between an Internet server and a user's computer
- **Peer-to-peer (P2P) file sharing**
  - Share data from one computer to another
  - Every user can be a server
  - Napster
    - Kazaa
    - Gnutella
  - With P2P, every user on the network can make *data* available to every other user on the network

Instant Messaging

- Allows user to create a private chat session with another user
- IM started with AOL
- IM sneaking into corporate networks
- Many Web-based companies use IM technology for customer service
  - State of Virginia
  - eBay

Usenet Newsgroups

- Online, *bulletin board* discussion forums
- Users post and read messages
- More than 100,000 newsgroups
- Millions of newsgroup readers
- Important information resource, especially for technical issues and products
- Newsgroup messages distributed using open standard
  - Many are uncensored

Streaming Audio & Video

- **Creating and sending audio and video files**
  - Sports
    - Men's intercollegiate basketball at sports.yahoo.com
    - Major league baseball
  - News
    - Fox News
    - CNN radio
  - Business
    - ZDNet
  - Education
    - Warriors of the Net

Internet Telephony

- Voice-over Internet Protocol (VoIP)
- Use your computer like a telephone
- Software connects computers via the Internet and transmits voice data
- Savings comes from eliminating toll charges between locations

The World Wide Web

- Collection of hyperlinked computer files on the Internet
- **Client-server application**
  - Web servers
  - Web browsers as clients
- **WWW standards**
  - **Hypertext markup language (HTML)**
    - Current standard for writing Web pages
    - Implementation of SGML specifically for Web pages
    - Tags in HTML instruct the client browser how to format and display the Web page content
  - **Hypertext transfer protocol (HTTP)**
    - **Protocol that establishes a connection between Web server and client**
  - **Extensible markup language (XML)**
    - A *meta*-markup language
    - Gives meaning to the data enclosed within XML tags

Static versus Dynamic Web Pages

- HTML and XML only display and exchange data
- Static: No interactivity; no processing of data
- **Dynamic**
  - **Scripting languages**
    - Provides basic interactivity
      - Rollovers
      - Crawling text
    - JavaScript
    - VBScript
- **Full-featured Web programming**
  - Java
  - Client side scripting or browser side scripting
  - Applets
  - J2EE
- **Common Gateway Interface (CGI)**
  - Allows passing of data between a static HTML page and a computer program

Figure 6.8 A Server Side Script

```
<!--- This query selects all members from the Students table in the irt
      datasource --->

<CFQUERY NAME="GetStudents" DATASOURCE="students">
   SELECT *
   FROM Students
</CFQUERY>

<html>
<head>
<title>Students of students.org</title>
</head>
<body>

<table border="1">
<tr>
<th>ID</th><th>Name</th><th>Age</th>
</tr>

<!--- Here's all students output nicely into a table --->

<CFOUTPUT QUERY="GetStudents">
<tr>
<td>#ID#</td><td>#Name#</td><td>#Age#</td>
</tr>
</CFOUTPUT>

</table>
</body>
</html>
```

Comments
HTML
Script

Result

ID	Name
11	Bob Smith
13	Joe Taylor
21	Ian Rogers
18	Kyle Scott

Source: tech.irt.org/articles/js123/#3

Searching the WWW

- Most data on the Internet is part of the WWW
- *Search engines* – large databases that index(فهرسه) WWW content
- Building the search engine database
  - Submit a site to the search engine administrator for listing
  - **Spiders**
    - **Metatags**
  - **Google**
  - **Yahoo**

Internet Governance

- **Internet Engineering Task Force (IETF)**
  - Works in groups to develop standards
- **Internet Engineering Steering Group (IESG)**
  - Approves or disapproves standards developed by the IETF
- **Internet Architecture Board (IAB)**
  - The oversight authority for the standards development process
- **World Wide Web Consortium (W3C)**
  - Promotes the WWW and develops new web technologies and standards

Intranets and Extranets

- **Intranet**
  - Internal company network that uses Internet and WWW technologies.
  - Flexible
  - Cost efficient
  - Uses
    - Find and share documents
    - Collaborate
    - Communicate
  - Only authorized users
    - Secured by firewall
- **Extranet**
  - Web site that allows customers and business partners limited access to organizational intranet
  - Uses Internet and WWW technologies
  - Add additional security through the use of a Virtual Private Network (VPN)

Impact of Internet and WWW

- Meredith – The Internet makes The Bead Bar a global company and expands market.
- Leda and Miriam– Use Web site to improve brand image and to create franchisee web sites.
- Mitch – Search for information on cruise industry and opportunities to leverage the Bead Bar.
- Julia – Pleased with low cost of Web site and reduced costs to connect headquarters with franchise locations.
- Rachel – Intranet will offer opportunity to reengineer purchasing procedures.
- Jim – Develop policies for employee training and also for appropriate use for business purposes.

Learning Goals Summary

In this chapter you have learned:

- ✓ The Internet's current architecture
- ✓ To identify and describe the major Internet applications
- ✓ The structure and definition of the World Wide Web
- ✓ How Internet and World Wide Web searches work
- ✓ How the Internet is governed
- ✓ What are intranets and extranets and how companies use them